

1 Carryover representation and inference for biomarker-treatment
2 interaction tests in aggregated N-of-1 trials under covariance
3 moderation: a factorial simulation comparison of nine analysis
4 specifications

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7 **Contents**

8	Abstract	2
9	1 Introduction	4
10	2 Simulation design	6
11	2.1 Aims	7
12	2.2 Trial designs	7
13	2.3 Data-generating mechanism	8
14	2.4 Estimand and analysis specifications	10
15	2.5 Choice of analysis specifications	13
16	2.6 Factorial grid and parameter settings	15
17	2.7 Inference and the cluster-robust correction	16
18	2.8 Performance measures and computing environment	17
19	3 Results	17
20	3.1 Headline performance table	18
21	3.2 Tier 1 grid overview	19
22	3.3 Matched-decay baseline	23
23	3.4 Power by analysis specification	23
24	3.5 Decay-form mis-specification	24
25	3.6 Type I error control and cluster-robust recalibration	25
26	3.7 Sensitivity analyses	27
27	3.8 A higher-precision check	34
28	4 Discussion	34
29	4.1 The lagged-treatment term does no work	34
30	4.2 The exposure-weighted advantage, and where it fails	35
31	4.3 What the robustness checks add	36
32	4.4 Dropout matters more than the method	37
33	4.5 What analysis is best in which situations, in practice	37
34	4.6 Scope, related work, and limits	39

35	5 Conclusions	41
36	6 Conflict of interest	43
37	7 Funding	44
38	8 Data availability statement	44
39	9 Reproducibility	44
40	10 References	46

41 Abstract

42 **Background.** Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to val-
43 idate predictive biomarkers.

- 44 • Under covariance moderation (an MVN differential-correlation architecture in which
45 the biomarker-treatment interaction lives in a treatment-state-dependent correlation),
46 a companion study found carryover costs up to roughly 31% of power in relative terms.
- 47 • The analyst must still decide how to represent residual exposure in the linear predictor.
- 48 • It is not established which representation is most robust to the ordinarily unknown
49 pharmacokinetic decay shape.

50 **Methods.** We compare nine analysis-model specifications for the drug-exposure regressor, in
51 two tiers.

- 52 • Tier 1 (primary evidence base): three specifications across a full factorial grid.
53 – Unadjusted: the binary on-drug indicator, ignoring carryover altogether.
54 – Lag-adjusted: augments that indicator with a lagged just-off-drug term estimating
55 carryover non-parametrically, the classical crossover device.
56 – Exposure-weighted: a continuous indicator committing to a parametric decay and
57 half-life, the current package default.
- 58 • These three are embedded in an ADEMP-structured factorial simulation¹ crossing two
59 data-generating decay forms (exponential, Weibull) with three trial designs, two sample
60 sizes, and three interaction strengths (216 cells, 500 replicates each), plus sensitivity
61 blocks and a cluster-robust recalibration.
- 62 • All three are fitted to the same simulated datasets (paired contrasts), and each perfor-
63 mance measure is reported with its Monte Carlo standard error (MCSE).
- 64 • Tier 2 extension: six further specifications at a single reference cell – an AIC-selected
65 half-life variant, a paired-difference regression with no repeated-measures model at all,
66 and CR2 cluster-robust variants of four of the above; a targeted comparison, not a full
67 grid.

68 **Results.** Three findings stand out.

- 69 • First, the lagged-treatment term buys nothing: Lag-adjusted and Unadjusted are sta-
70 tistically indistinguishable across all 216 Tier 1 cells, differing by a mean of 0.003 in

power against an MCSE of 0.018, and agreeing to three decimal places on bias, MSE, and coverage – a clean null, since the two share an estimand and differ by exactly one nuisance term.

- Second, Exposure-weighted attains the highest Tier 1 power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), an advantage of about 10 percentage points (0.860 vs 0.766 for Unadjusted at the reference cell).
 - Under the classical crossover (CO) design it is markedly inferior (0.488 vs 0.830).
 - Where it leads, the lead is robust to half-life mis-specification and to autocorrelation strength, and widens under a light-tailed decay-shape mis-specification, but narrows to statistical insignificance under the two most heavy-tailed Weibull shapes examined.
 - The model-based interaction test is mildly conservative for all three (mean null rejection 0.036 to 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking.
- Third, at the Hybrid reference cell the Tier 2 extension finds that combining the AIC-selected half-life with a CR2 standard error outperforms every other specification tested, including the CR2-corrected Exposure-weighted predictor, with Type I error held at nominal despite the post-selection risk the half-life selection step raises in principle.
 - This result held at only that one cell, reported as a promising direction rather than a validated recommendation.

Conclusions. Four points warrant emphasis.

- First, under covariance moderation we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs; under a crossover design it should not be used, and the unadjusted binary indicator is materially better.
- Second, the classical lagged-treatment remedy cannot be recommended in any design examined here, since it never improved on ignoring carryover altogether – traced to its entering the model as a nuisance main effect, which leaves the attenuation of the interaction untouched.
- Third, across all conditions anticipated dropout governs power far more than the choice among specifications, and should dominate design effort accordingly.
- Fourth, the AIC-selected half-life specification with a CR2 correction merits further validation as a candidate default, having outperformed every other specification at the one cell where all nine were compared directly, but it has not yet been tested against the same robustness battery the Tier 1 three have.
- This manuscript restricts the grid to the covariance-moderation architecture and to the exponential and Weibull decay forms; the mean-moderation architecture and linear decay are evaluated in the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

Section summary:

- The manuscript compares nine analysis-model specifications (Tier 1: three; Tier 2: six more at one reference cell) for representing drug exposure/carryover, restricted to covari-

ance moderation, using a 216-cell, 500-replicate-per-cell ADEMP-compliant simulation.

- Three headline results: the lagged-treatment term is useless; exposure weighting wins by ~10 points in Hybrid/OL+BDC but loses badly under crossover; and a Tier-2 AIC-selected-half-life-plus-CR2 specification beats everything at one cell, provisionally.
- The Conclusions recommend exposure weighting (with CR2) only for Hybrid/OL+BDC under covariance moderation, reject the lagged-treatment remedy outright, flag dropout as the dominant practical lever, and flag the AIC+CR2 specification as promising but not yet fully validated.

1 Introduction

- In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator; the within-patient on/off contrast estimates that patient's individual treatment effect.
- An aggregated N-of-1 trial pools many such series into a mixed model, trading within-patient resolution for population-level inference at sample sizes considerably smaller than a parallel-group trial would require².
- The estimand is the biomarker-treatment interaction: whether a baseline covariate B_i modifies the treatment effect.
- Three trial designs, differing in how on-drug/off-drug occasions are arranged, are examined: classical crossover (CO), open-label followed by blinded discontinuation (OL+BDC), and a Hybrid N-of-1 design (each defined in Section 2.2).
- Carryover is the inferential complication specific to this setting: a drug's pharmacological effect decays over days or weeks rather than switching off abruptly, so nominally off-drug observations still carry residual exposure.
- An exposure regressor ignoring this is mis-measured, and the damage is twofold:
 - The interaction contrast is diluted, since occasions still carrying substantial drug effect are coded identically to occasions carrying none, biasing the biomarker-by-exposure coefficient toward zero (as any error-prone covariate would).
 - Those same occasions are heterogeneous in a way the model does not represent, so unexplained variation inflates the standard error of the already-attenuated coefficient.
- The test loses on both counts, and power falls accordingly.
- This is a loss of power, not of validity: under the null ($c_{bm} = 0$), no biomarker-response association exists at any level of residual exposure, so a mis-coded regressor has no interaction to distort and test size is unaffected by exposure coding.
- The mild conservatism observed in Section 3.6 has a separate origin in the working covariance and applies equally across specifications; carryover mis-specification is therefore a question of efficiency, and the study is framed in terms of recovered power.
- Three answers are available for representing residual exposure, corresponding to three analyst positions toward the unknown decay: deny it (strictly on/off coding), estimate it (add a nuisance term whose magnitude the data determine), or assume it (commit to a decay curve and half-life).

- 154 • A companion study³, building on Hendrickson², shows that under covariance moderation
155 ($\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$), carryover erodes power directly and design-dependently:
156 at a half-life of one week, the relative loss is roughly 31% under OL+BDC but only 3
157 to 5% under crossover and Hybrid, against 1 to 3% under an alternative architecture
158 where the biomarker moderates the mean response directly.
- 159 • Attention is restricted here to covariance moderation, where the carryover problem is
160 largest and Hendrickson's original results were obtained; that earlier work fixed the
161 decay shape at exponential and evaluated only one analysis specification, leaving open
162 which residual-exposure representation performs best when the true decay shape is not
163 known precisely (the ordinary case).
- 164 • Three specifications form the primary evidence base:
 - 165 – Unadjusted: the binary on-drug indicator, ignoring carryover entirely, serving as
166 the benchmark any remedy must beat.
 - 167 – Lag-adjusted: augments it with a lagged just-off-drug term estimating carryover
168 magnitude without assuming a functional form, following the classical crossover-
169 trial device of Jones and Kenward⁴.
 - 170 – Exposure-weighted: a continuous indicator committing to a parametric decay form
171 and half-life, the current pmsimstats-ng default.
- 172 • Unadjusted and Lag-adjusted estimate the same coefficient and are fitted to the same
173 simulated datasets, differing by exactly one nuisance term, so their contrast isolates
174 that term's contribution alone; Exposure-weighted differs on a second axis too, since it
175 changes the regressor carrying the interaction.
- 176 • All three are crossed against exponential and Weibull data-generating decay forms, ask-
177 ing whether one specification can be recommended outright or whether the choice de-
178 pends on trial design, following the ADEMP reporting framework of Morris, White, and
179 Crowther¹ throughout.
- 180 • Six further specifications (an AIC-selected half-life variant, a paired-difference regression,
181 and CR2 cluster-robust variants of four base specifications) are added later at a single
182 reference cell (Section 2.5), once the Tier 1 results raise specific new questions the grid
183 alone cannot answer.
- 184 • Closing roadmap: Section 2 describes the simulation design in ADEMP order, Section
185 3 gives the results, Section 4 takes up the practical implications.

186 Section summary:

- 187 • Carryover degrades interaction-test power in aggregated N-of-1 trials through two dis-
188 tinct mechanisms (attenuation and inflated residual variance), without biasing the null
189 under $c_{bm} = 0$.
- 190 • A companion study shows carryover's relative power cost is sharply design-dependent
191 (~31% under OL+BDC vs 3-5% under CO/Hybrid, vs 1-3% under an alternative archi-
192 tecture), motivating this manuscript's restriction to covariance moderation.
- 193 • Three analyst positions toward unknown decay (deny/estimate/assume) motivate the
194 three Tier-1 specifications (Unadjusted, Lag-adjusted, Exposure-weighted), with six fur-
195 ther Tier-2 specifications added later to answer questions the Tier 1 grid alone cannot.

2 Simulation design

- The study simulates complete aggregated N-of-1 trials and analyzes each one under every specification.
- Single-replicate procedure:
 - Patients are allocated across the randomization paths of the chosen trial design (Section 2.2).
 - For each patient, a pre-treatment biomarker plus three latent response components (pharmacological response, natural-history trend, placebo-belief) are drawn jointly across all eight measurement occasions from a multivariate normal distribution.
 - Component means follow modified Gompertz trajectories.
 - Covariance carries within-component AR(1) autocorrelation, cross-component correlations, and the treatment-state-dependent biomarker-response correlation that encodes the interaction (Section 2.3).
 - Observed symptom score at each occasion = baseline severity minus the sum of the three components (a beneficial effect lowers the score).
 - Carryover enters through a decay function of time since discontinuation, keeping part of the pharmacological effect present at nominally off-drug occasions.
 - All three Tier-1 analysis specifications (Section 2.4) are then fitted to that same simulated dataset, recording the interaction coefficient, SE, and p -value from each.
 - Repeated 500 times for every parameter combination in the grid (Section 2.7).
- Fitting all specifications to a common dataset is deliberate:
 - Makes every contrast among them paired.
 - Differences are estimated with substantially less Monte Carlo variance than independent draws would give.
- N denotes the **total** number of patients in a trial, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P patients to each.
- Worked example: at $N = 70$ this is 35 per path under the two-path designs and 17 or 18 per path under the four-path Hybrid design.

Section summary:

- Each simulated replicate draws a biomarker and three latent response components jointly from an MVN structure with AR(1) autocorrelation, cross-component correlations, and a treatment-state-dependent biomarker-response correlation encoding the interaction; carryover is applied through a post-discontinuation decay function.
- All specifications are fit to the same simulated dataset (common random numbers), sharpening pairwise contrasts.
- N is always the total sample size across paths, allocated evenly; this definitional point governs how per-path sample sizes should be read throughout the manuscript.

2.1 Aims

- The aim is to quantify, under covariance moderation, the cost of mis-specifying the carryover representation in the analysis model, and to determine which of the three candidate strategies performs better across the grid of data-generating decay forms and trial designs.
- A second aim, addressed in the Tier 2 sensitivity blocks of Section 3.7, is to check whether the resulting ranking survives perturbation of within-subject autocorrelation, dropout rate, half-life accuracy, and interaction size.

Section summary:

- The study has two aims: rank the three carryover-handling strategies under covariance moderation across decay forms and trial designs, and stress-test that ranking's robustness in the Tier 2 sensitivity blocks.

2.2 Trial designs

- The three designs differ in how on-drug/off-drug occasions are arranged, and so in how much post-discontinuation information each generates.
- All three schedule eight measurement occasions per patient across randomization paths, each a fixed on-drug/off-drug schedule.
- Each design carries an expectancy weight (one during open-label periods, one half during blinded periods) scaling the placebo-belief component.
- **Crossover (CO).** A traditional two-period within-patient crossover, blinded throughout, occasions spaced 2.5 weeks apart from 2.5 to 20 weeks.
 - One path is active for the first four occasions then comparator.
 - The other path reverses that order and so never discontinues, contributing no post-discontinuation observations.
- **Open-label followed by blinded discontinuation (OL+BDC).** Open-label titration at 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo.
 - The two paths discontinue after week 17 or 18, so only two or three occasions follow.
- **Hybrid.** The Hendrickson design and this program's reference.
 - Open-label titration at weeks 4 and 8, followed by blinded discontinuation weekly from week 9 to 12, then a brief two-occasion crossover at weeks 16 and 20.
 - The four paths vary discontinuation timing and crossover-arm order.
 - It is the richest of the three in on-off transitions and the only one allowing more than one discontinuation per patient.
- Practical consequence: the first post-discontinuation occasion falls one week after switching under OL+BDC and Hybrid, but 2.5 weeks under CO.
- At the half-lives examined, roughly half the pharmacological effect persists into the first off-drug occasion under the two non-crossover designs, against less than a fifth under CO, so the designs differ substantially in how much carryover contamination they can accumulate at all (Section 4.2).

275 Section summary:

- 276 • The three trial designs (CO, OL+BDC, Hybrid) share an 8-occasion structure but differ
277 sharply in schedule and in how soon after discontinuation the first off-drug measurement
278 occurs.
- 279 • That timing difference (1 week vs 2.5 weeks) drives a large difference in how much
280 residual carryover effect remains at the first off-drug occasion (roughly half vs under a
281 fifth), foreshadowing the design-dependent results in Section 3.4 and beyond.

282 2.3 Data-generating mechanism

- 283 • Under covariance moderation, the biomarker and biological response are drawn jointly
284 from a multivariate normal distribution with a treatment-state-dependent correlation,
285 $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} the time since drug
286 discontinuation.
- 287 • Mean moderation and the combined dual-channel architecture fall outside this
288 manuscript's scope; the full cross-architecture comparison is in the archived full-scope
289 drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`).
- 290 • The decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ maps t_{sd} to a residual-effect multiplier; two forms
291 are evaluated here, each matched to a common half-life $t_{1/2}$.
- 292 • A third form, linear decay, appears in the full-scope manuscripts but not here.

293 2.3.1 Exponential decay

$$294 \quad \phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

- 295 • This is the decay form used throughout the companion manuscript.
- 296 • Implemented in `functions.R` under `implementations/tidyverse/`.
- 297 • Corresponds to classical first-order elimination kinetics, the default pharmacokinetic
298 assumption.
- 299 • Serves as the reference form against which Weibull decay is compared.

300 2.3.2 Weibull decay

$$301 \quad \phi_{\text{weib}}(t_{sd}) = \exp\left(-(\lambda_w t_{sd})^k\right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

- 302 • Shape parameter k governs tail behavior:
 - 303 – $k < 1$: heavier (slower) elimination tail than exponential.
 - 304 – $k > 1$: lighter (faster) elimination tail than exponential.
 - 305 – $k = 1$: recovers the exponential form exactly.
- 306 • Four values reported: $k \in \{0.25, 0.5, 2.0, 4.0\}$, two on each side of exponential.
 - 307 – $k = 0.5$ and the more extreme $k = 0.25$: progressively more prolonged low-
308 concentration tails.
 - 309 – $k = 2.0$ and the more extreme $k = 4.0$: progressively faster clearance once concen-
310 tration falls below some threshold.

- This range is wider than an earlier version of the grid, which used only $k = 0.7, 1.5$.
- $k = 1.0$ is not reported as a fifth Weibull condition: it is numerically indistinguishable from Exponential, giving five decay-shape levels in the grid (not six).

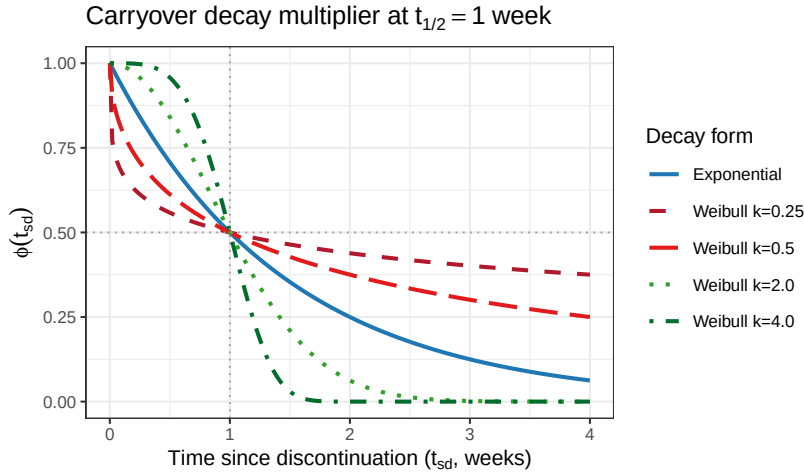


Figure 1: The carryover decay multiplier $\phi(t_{sd})$ at the reference half-life ($t_{1/2} = 1.0$ week) for exponential decay and the four Weibull shapes. The dotted reference lines mark the defining half-life point ($\phi = 0.5$ at $t_{sd} = t_{1/2}$), which all five curves pass through by construction; the shape parameter governs only how quickly ϕ falls before that point and how slowly it decays after.

- Figure 1 shows the shape difference concretely.
- $k = 0.25$ and $k = 0.5$ curves have progressively heavier tails, staying well above the exponential curve for every $t_{sd} > t_{1/2}$, most markedly at $k = 0.25$.
- Practical consequence: the analysis-side predictor picks up substantially more residual carryover during a prolonged washout than the half-life alone suggests, increasingly so as k falls further below one.
- $k = 2.0$ and $k = 4.0$ curves have progressively lighter tails, clearing sharply faster once concentration falls below the half-maximum threshold.
- All five curves pass through the same half-life-defining point ($\phi = 0.5$ at $t_{sd} = 1.0$), so the five forms remain comparable at a fixed nominal half-life despite diverging shapes on either side of it.

Section summary:

- Two decay forms are used: exponential (the companion manuscript's form, first-order elimination kinetics) and Weibull, with a shape parameter k controlling tail weight relative to exponential.
- Five decay-shape levels are examined in total (exponential plus four Weibull shapes, $k \in \{0.25, 0.5, 2.0, 4.0\}$); $k = 1.0$ is excluded as duplicating exponential.
- All five decay curves are calibrated to the same half-life-defining point, so comparisons across shape are not confounded by a shifted half-life.

Table 1: All nine analysis specifications evaluated in this manuscript. G1-G3 are the Tier 1 headline three (Section 2.4); G4-G5 are Tier 2 additions replacing the retired E4/E5/E6 (Blocks S7-S9, Section 3.7); G6-G9 are CR2 cluster-robust variants of G1, G2, G4, and G3 respectively (Block S10). G10 (G5+CR2) does not apply and is omitted: G5 collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

2.4 Estimand and analysis specifications

- The interaction estimand is the on-drug biomarker-response correlation c_{bm} , a dimensionless quantity.
- Bias and empirical SE are reported on a calibrated effect-size scale (expected on-drug response difference per SD of biomarker at $t_{1/2} = 0$), per docs/19-mean-moderation-implementation-note
- Power and Type I error are reported on the native test-statistic scale, with null estimand $c_{bm} = 0$.
- Shared model formula for the Tier 1 three: `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))`, differing only in how X is constructed and whether a nuisance term is added.
- Unadjusted and Lag-adjusted both test $\text{bm}:D_{it}$, so every performance measure is comparable between them.
- Exposure-weighted tests a different quantity, $\text{bm}:D_{bc,it}$.
 - Power and Type I error remain comparable across all three specifications (properties of the test).
 - Bias, MSE, and coverage are **not** comparable across that divide (cross-referenced to Section 3.1).
- Nine specifications total are evaluated across the manuscript: the Tier 1 three, plus six Tier 2 additions (introduced in the next section, tested in Blocks S7-S10, Section 3.7).
- Internally (code and data files, Reproducibility section), each specification is stored under an E-prefixed code (E1, E2, ...); the manuscript's prose refers to all nine uniformly by the compact G1-G9 code (defined in `spec-labels.R`) or by name, per Table 1 below.

2.4.1 Unadjusted (G1): binary on-drug indicator

- $X = D_{it}$, the binary on-drug indicator (one on drug, zero off).
- Carryover is not represented at all.
- An off-drug occasion still carrying most of the pharmacological effect is coded identically

359 to one carrying none.

- 360 • This is the default an analyst obtains without considering carryover; treated as the
361 benchmark other specifications must justify their added structure against.
- 362 • Consumes no half-life; invariant to the analyst's pharmacokinetic assumptions.

363 2.4.2 Lag-adjusted (G2): binary indicator plus estimated carryover term

- 364 • Retains $X = D_{it}$; adds a lagged-treatment indicator, $L_{it} = D_{i,t-1}(1 - D_{it})$, marking
365 off-drug timepoints immediately following an on-drug one.
- 366 • This is the classical crossover-trial device of Jones and Kenward⁴.
- 367 • Carryover magnitude is estimated from the data, not assumed; like Unadjusted, this
368 specification is invariant to the assumed half-life.
- 369 • L_{it} enters only as a nuisance main effect; there is no $\text{bm}:L_{it}$ product, so the interaction
370 actually tested remains $\text{bm}:D_{it}$.
- 371 • Lag-adjusted can absorb mean-level carryover but cannot repair the interaction's atten-
372 uation (Section 1).
- 373 • Framed as Unadjusted with a carryover nuisance control added, not a fundamentally
374 different model of the interaction (Section 4.1).

375 2.4.3 Exposure-weighted (G3): continuous decayed predictor

- 376 • $X = \text{Dbc}_{it}$, a continuous exposure-decayed predictor matched to the analyst's assumed
377 decay form and half-life; the current pmsimstats-ng package default.
- 378 • At on-drug timepoints, $D_{bc,it} = 1$.
- 379 • Off drug it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing half its value every $\hat{t}_{1/2}$ weeks after
380 discontinuation.
- 381 • When the assumed decay form matches the true one, the predictor is well matched;
382 when it does not, it is mis-specified, probed directly in Block S2.
- 383 • The half-life assumption enters only through the predictor values handed to `lme`, not
384 the model formula, so this specification is in effect a different model for each assumed
385 half-life.
- 386 • It is the only one of the three original specifications that consumes a half-life at all; the
387 S2 sensitivity contrast (Section 3.7) is designed to probe this.
- 388 • Limiting relationship: as $\hat{t}_{1/2} \rightarrow 0$, $D_{bc,it} \rightarrow D_{it}$, so Unadjusted is exactly the boundary
389 member of the Exposure-weighted family at zero assumed half-life.
- 390 • The S2 sweep is therefore a traverse anchored at Unadjusted, not an independent check.

391 2.4.4 AIC-selected half-life (G4)

- 392 • Uses Exposure-weighted's formula, $X = \text{Dbc}_{it}$, but chooses $\hat{t}_{1/2}$ from the data rather
393 than assuming it.

$$394 \quad \hat{t}_{1/2} = \arg \min_{t_{1/2} \in \mathcal{T}} \text{AIC}(M(t_{1/2})), \quad \mathcal{T} = \{0.25, 0.5, 1.0, 2.0\} \text{ weeks},$$

- $AIC(M) = -2 \log L(M) + 2p$, $L(M)$ the fitted model's likelihood and p its number of estimated parameters⁵.
- Reported power, coefficient, and SE are read from $M(\hat{t}_{1/2})$ alone, the single winning fit, not pooled or averaged across $\mathcal{T}$.
- Mirrors `characterize_carryover()` (R/`carryover_analysis.R`), the package's own real-data half-life-selection routine.
- Like Exposure-weighted, targets $bm:D_{bc,it}$; unlike Exposure-weighted, requires no half-life from the analyst at all.
- Cost: a post-selection inference risk⁶ – $M(\hat{t}_{1/2})$'s reported SE and p -value treat $\hat{t}_{1/2}$ as fixed in advance, when the same data in fact chose it, so nominal inference does not formally account for the selection step.
- Block S11 (Section 3.7) checks whether this risk manifests as Type I error inflation in practice.

2.4.5 Paired-difference (G5)

- A two-stage, correlation-structure-agnostic alternative, not an elaboration of the mixed-model family above.
- Stage 1: for each patient i with $n_i^{\text{on}} > 0$ on-drug and $n_i^{\text{off}} > 0$ off-drug occasions, define per-patient on-drug and off-drug means:

$$\bar{S}_{x,i}^{\text{on}} = \frac{1}{n_i^{\text{on}}} \sum_{t: D_{it}=1} S_{x,it}, \quad \bar{S}_{x,i}^{\text{off}} = \frac{1}{n_i^{\text{off}}} \sum_{t: D_{it}=0} S_{x,it},$$

- and their difference $\Delta_i = \bar{S}_{x,i}^{\text{on}} - \bar{S}_{x,i}^{\text{off}}$.
- Stage 2: OLS regression of Δ_i on centered biomarker across $N' \leq N$ patients contributing a difference:

$$\Delta_i = \beta_0 + \beta_1 bm_i + \varepsilon_i, \quad i = 1, \dots, N',$$

- Interaction test is on β_1 .
- No repeated-measures model, no carryover representation, no correlation structure of any kind: Δ_i collapses each patient's eight occasions to one number before bm ever enters.
- Framed as a direct biomarker-interaction extension of the paired t-test that Senn, Julious and Araujo⁷ found to outperform carryover-adjusted mixed models for ordinary N-of-1 treatment-effect estimation.
- Patients contributing only on-drug or only off-drug observations (e.g., CO's never-discontinuing path) have $n_i^{\text{on}} = 0$ or $n_i^{\text{off}} = 0$, cannot form Δ_i , and are dropped, so $N' < N$ whenever such a path exists in the design.

2.4.6 CR2 cluster-robust variants (G6-G9)

- Section 3.6 (Block S6) shows the model-based interaction test is mildly conservative, and a CR2 bias-reduced cluster-robust SE restores nominal size without disturbing the

ranking among G1-G3.

- Uses the estimator of Bell and McCaffrey⁸, via `clubSandwich`, paired with Satterthwaite degrees of freedom.
- That correction (`lme+CR2` in Section 2.7) was applied only to G3 there; Block S10 (Section 3.7) extends it to G1, G2, and G4 as well, replacing only the SE, reference distribution, and p -value on each specification's existing point estimate.
- Mapping: G6 (G1+CR2), G7 (G2+CR2), G9 (G4+CR2).
- G8 (G3+CR2) is the same specification already introduced in Section 2.7 and Block S6, carried into the nine-way comparison at Block S10 without rerunning it.
- A tenth combination, G5+CR2, does not apply: G5 already collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

Section summary:

- Nine analysis specifications (G1-G9) are ultimately compared, built from three families: the Tier 1 three (Unadjusted, Lag-adjusted, Exposure-weighted), two Tier 2 additions (G4 AIC-selected half-life, G5 paired-difference), and CR2 cluster-robust variants (G6-G9) of the mixed-model specifications.
- Unadjusted/Lag-adjusted share an estimand ($\text{bm}:D_{it}$) and so are directly comparable on every measure; Exposure-weighted, G4, G9 target a different estimand ($\text{bm}:D_{bc,it}$), so only power and Type I error are comparable across that divide.
- G5 discards the mixed-model framework entirely in favor of a two-stage paired-difference regression, motivated by prior evidence that simplicity is not obviously costly in N-of-1 analysis.
- Internally all nine specifications are stored under E-prefixed codes; the manuscript displays them uniformly as G1-G9 via `spec-labels.R`.

2.5 Choice of analysis specifications

- Three considerations bounded which specifications were tested, rather than surveying the wider N-of-1/crossover-trial literature.
- The Tier 1 three were chosen because each represents a distinct, established position an analyst can take toward unmeasured/assumed decay, not because they exhaust the space:
 - Unadjusted: the default any analyst obtains without considering carryover at all; necessary benchmark.
 - Lag-adjusted: follows Jones and Kenward's⁴ classical crossover-trial device.
 - Exposure-weighted: the `pmsimstats-ng` package's own established default, so the practical stakes of getting its behavior right are correspondingly higher than for the other two.
- Untested alternative model classes, left out by design:
 - Bayesian hierarchical N-of-1 models.
 - GEE and other marginal models (compared systematically, not just as a robustness check, per Section 2.7).
 - Randomization and permutation tests.

- 472 – Functional or spline-based flexible decay curves.
- 473 – State-space or Kalman-filter treatments of drug concentration.
- 474 – Two-stage “series of N-of-1 trials” meta-analytic combination.
- 475 • This is a scoping decision, not an oversight: the manuscript’s aim is to compare carry-
476 over *representations* within a single, fixed conditional mixed-model framework matching
477 Hendrickson’s² original analysis model, not a comprehensive methods bake-off (future
478 work: Section 4.6).
- 479 • Blocks S7-S9 (Section 3.7) revisit this scoping decision once, for a specific reason: Section
480 4.1 shows no carryover term tested so far can repair the interaction’s attenuation as long
481 as it enters only as a nuisance main effect; the natural next model interacts the carryover
482 term with the biomarker directly.
- 483 • An earlier version of Block S7 tested two such interacted specifications:
484 – E5: a lagged term crossed with **bm**.
485 – E6: a continuous elapsed-time term crossed with **bm**.
- 486 • Both underperformed Unadjusted at every cell examined, across three blocks, two trial
487 designs, two architectures, and three carryover half-lives (Sections 3.6, 4.5).
- 488 • Traced to a structural cause: the DGP has a single true decay parameter, so crossing a
489 second carryover term with the biomarker adds a free parameter with no counterpart in
490 the truth being estimated; two collinear regressors chasing one true degree of freedom
491 lose power to a correctly specified single-parameter model almost by construction.
- 492 • A further variant of the same interacted-term family is expected to fail on the same
493 structural argument, so E5/E6 were replaced by two specifications answering different
494 open questions about the analyst’s unknown true decay half-life:
495 – G4 asks whether estimating the half-life from the data recovers the power a correctly
496 assumed half-life would give; selects by AIC over a candidate grid, reusing the logic
497 of `characterize_carryover()` (R/`carryover_analysis.R`), already used for real-
498 data analysis; addresses Block S2’s question from the opposite direction (estimation
499 instead of assumption).
- 500 – G5 asks a more basic question raised by the wider literature: is a conditional mixed
501 model even necessary? Senn, Julious and Araujo⁷ compared four analysis strate-
502 gies for ordinary N-of-1 trials and found a simple paired t-test on within-patient
503 treatment differences achieved near-nominal Type I error, higher power, and com-
504 parable bias/MSE relative to a carryover-adjusted mixed model, recommending
505 the mixed model only when carryover is specifically suspected. G5 is a direct
506 biomarker-interaction extension of their preferred method.
- 507 • This remains a bounded comparison: G4 and G5 were chosen because each is imple-
508 mentable within, or trivially adjacent to, the existing analysis pipeline (Section 2.7),
509 and each answers a specific question the manuscript’s own results raised, not because
510 they close the wider literature gap.
- 511 • Bayesian, GEE-as-primary-model, randomization-test, and two-stage meta-analytic al-
512 ternatives remain untested and are discussed as future work (Section 4.6).

513 Section summary:

- The Tier 1 three specifications were chosen as distinct, established analyst positions, not as an exhaustive survey of the N-of-1 literature; a wide range of model classes was left untested by deliberate scoping decision.
- An earlier interacted-carryover-term family (E5/E6) failed structurally because the DGP has a single true decay parameter, and was replaced by G4 (AIC-selected half-life) and G5 (paired-difference regression), each targeting the analyst's unknown-half-life problem from a different angle.
- G4 and G5 remain a bounded addition, not a comprehensive methods comparison; several major alternative frameworks (Bayesian, GEE, randomization tests, two-stage meta-analysis) are explicitly deferred to future work.

2.6 Factorial grid and parameter settings

- The principal comparison crosses biomarker moderation, carryover half-life, trial design, and sample size against the three Tier-1 analysis specifications, under covariance moderation and the exponential DGP.

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (covariance moderation) only
Replicates per cell	500 (target), 50 for development

- Three specifications, three half-lives, three designs, two sample sizes, three biomarker levels: $3 \times 3 \times 3 \times 2 \times 3 = 162$ cells.
- The $c_{bm} = 0$ row is the Type I error check against nominal 5%.
- The $t_{1/2} = 0$ row (no carryover) gives a best-case power reference at each combination of N , design, and c_{bm} , against which the cost of nonzero carryover can be read off directly.
- Decay-shape sensitivity (Figures 3 and 5) is evaluated separately, at the reference cell only ($c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks), not across the full grid:
 - Exponential decay and four Weibull shapes ($k \in \{0.25, 0.5, 2.0, 4.0\}$, Section 2.3) crossed with three specifications, three designs, two sample sizes give $5 \times 3 \times 3 \times 2 = 90$ further cells (23-run-decay-shape-sensitivity.R).
- Rationale for restricting the decay-shape axis to one cell: it is the axis every other result in this manuscript holds fixed at exponential, and a full crossing would multiply simulation cost several times over for a comparison used in only two figures.
- $k = 1.0$ is not a fifth Weibull condition anywhere in this manuscript: it recovers the exponential form exactly (Section 2.3).

Section summary:

- The principal factorial grid has 162 cells crossing c_{bm} , $t_{1/2}$, N , and trial design against the three Tier-1 specifications, plus a separate 90-cell decay-shape sensitivity grid evaluated

only at the reference cell.

- The $c_{bm} = 0$ row anchors Type I error checks; the $t_{1/2} = 0$ row anchors the best-case (no-carryover) power reference used throughout the Results.

2.7 Inference and the cluster-robust correction

- The interaction test described so far is model-based: the SE of $\beta_{bm:X}$ is read from the fitted `lme` object, valid only if the assumed residual covariance is correct.
- This assumption is not innocuous: the working covariance (patient random intercept plus `corCAR1` AR(1) residual correlation) does not match the covariance the DGP actually induces⁹, inflating the SE by roughly six to ten percent and depressing the rejection rate correspondingly.

$$\kappa = \frac{\overline{SE}}{SD(\hat{\beta}_{bm:X})} = \frac{\frac{1}{n_{sim}} \sum_{r=1}^{n_{sim}} \widehat{SE}_r}{\sqrt{\frac{1}{n_{sim}-1} \sum_{r=1}^{n_{sim}} (\hat{\beta}_{bm:X,r} - \hat{\beta}_{bm:X})^2}},$$

- κ is the mean model-reported SE across replicates divided by the empirical SD of replicate-level point estimates.
 - $\kappa = 1$: correctly calibrated.
 - $\kappa > 1$: model overstates uncertainty (a conservative test).
- At the null cell, κ runs 1.05 to 1.10 across specifications, with empirical Type I error 0.029 to 0.039 against nominal 0.05.
- Remedy: a cluster-robust (sandwich) variance estimator, valid under covariance misspecification, deriving variance from observed between-cluster variation rather than the assumed covariance.
- Clusters are patients (eight occasions each, unique across randomization paths).

$$\widehat{\text{Var}}_{\text{sandwich}}(\hat{\beta}) = \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} X_i \right)^{-1} \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} \hat{e}_i \hat{e}_i' \hat{V}_i^{-1} X_i \right) \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} X_i \right)^{-1},$$

- This naive (Liang-Zeger) sandwich estimator substitutes each patient's own observed $\hat{e}_i \hat{e}_i'$ for the single assumed $\hat{V}_i$, so a mismatched working covariance no longer biases the variance estimate – at the cost of needing enough clusters for stability.
- With only $N = 35$ or 70 clusters, the naive sandwich form is itself markedly downward-biased, so the CR2 bias-reduced estimator of Bell and McCaffrey⁸ is used instead.
- CR2 replaces each raw residual $\hat{e}_i$ with an adjusted residual $\hat{e}_i^{\text{CR2}} = A_i \hat{e}_i$, chosen so the recomputed sandwich variance is exactly unbiased under the working model (full construction of A_i in Bell and McCaffrey⁸, not reproduced here).
- Paired with Satterthwaite degrees of freedom (both via `clubSandwich`); replaces only SE, reference distribution, and p -value, leaving point estimates untouched.
- The correction is applied in two places:
 - A null-cell diagnostic ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates) establishing whether CR2 restores nominal size (Section 3.6).
 - A five-cell block (reference cell plus four stress cells: small N , high autocorrelation,

half-life mis-specification, 30% MCAR dropout; 500 replicates each; labeled S6), asking whether restoring size disturbs the ranking (same section).

- Blocks S1-S4 are the Tier 2 sensitivity analyses of Section 3.7.
- Block S5 (exploratory autocorrelation-by-carryover interaction) is outside the present scope, reported only in the archived full-scope drafts.

Section summary:

- The default model-based interaction test is mildly conservative because the working covariance does not match the true data-generating covariance, quantified by the calibration ratio κ (1.05-1.10 at the null cell).
- A CR2 bias-reduced cluster-robust SE (Bell and McCaffrey), paired with Satterthwaite degrees of freedom, is the chosen remedy, applied at a null-cell diagnostic and a five-cell stress block (S6).
- Blocks S1-S5 constitute the remaining Tier 2 sensitivity analyses; S5 is out of scope here and reported only in archived drafts.

2.8 Performance measures and computing environment

- Reported per cell: Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al.¹, Tables 5-6.
- $n_{sim} = 500$ chosen so the MCSE on power at $p = 0.5$ would not exceed 0.022.
- No simulation runs for this manuscript at render time; it filters the same 540-cell production grid summary used in the archived full-scope drafts down to the 216-cell slice defined by covariance moderation and the two non-linear decay forms.
- All three specifications are fit to the same simulated dataset within each cell (common random numbers), so every pairwise contrast – especially Lag-adjusted minus Unadjusted – carries reduced Monte Carlo variance relative to independent draws.
- The Tier 2 sensitivity blocks and S6 recalibration are filtered from the same shared production outputs.
- Exact file paths and versions are given in the Reproducibility section.

Section summary:

- Performance is reported on the standard Morris et al. ADEMP measure set, with $n_{sim} = 500$ calibrated to an MCSE ceiling of 0.022 on power.
- This manuscript performs no new simulation at render time; it reads and filters existing production outputs, relying on common random numbers across specifications to sharpen pairwise contrasts.

3 Results

- All reported results derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to the covariance-moderation architecture and the exponential and Weibull decay forms.

Table 2: Headline performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$, the primary outcome of this manuscript. Bias is $\text{mean}(\text{estimate} - \theta_{\text{true}})$ on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration; see the estimand-mismatch caveat above for Unadjusted/Lag-adjusted). Empirical SE is the standard deviation of the point estimates across replicates, the actual sampling variability of the estimator, as distinct from whatever SE any single fit reports (the comparison the calibration ratio κ of Section 2.7 is built on). Coverage is the proportion of replicates whose nominal 95% Wald CI contains θ_{true} ; the low Unadjusted/Lag-adjusted values reflect the same estimand mismatch as Bias, not miscalibrated inference. MSE is $\text{bias}^2 + \text{Empirical SE}^2$, an overall accuracy summary. Non-conv is the fraction of replicates where ‘nlme::lme()’ failed to converge (zero throughout at this cell). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Lag-adjusted	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Exposure-weighted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	Unadjusted	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	Lag-adjusted	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
0.5	Exposure-weighted	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
1.0	Unadjusted	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	Lag-adjusted	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000
1.0	Exposure-weighted	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000

- The Monte Carlo SE on power does not exceed 0.022 at $p = 0.5$; differences smaller than this should not be over-interpreted.
- Coverage of the nominal 95% Wald CI is reported for the reference cell.
- Non-convergence rates are reported per cell in the tables that follow.

Section summary:

- Every Results figure and table is drawn from a single 500-replicate production run (covariance moderation, exponential/Weibull decay only), with an MCSE ceiling of 0.022 on power that should be used to judge whether any reported gap is meaningful.

3.1 Headline performance table

- Table 2 reports performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout.
- Bias, MSE, and coverage columns are computed against Exposure-weighted’s calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$.
- Unadjusted and Lag-adjusted estimate a different coefficient, $\text{bm}:D_{it}$; their apparent bias and under-coverage reflect this estimand mismatch, not an estimator defect.
- Those columns are comparable within the Unadjusted/Lag-adjusted pair but not across to Exposure-weighted.
- For the three-way comparison, power and Type I error (properties of the test rather than the coefficient) are the measures to rely on.

Section summary:

- The headline table anchors the manuscript’s reference cell (exponential, Hybrid, $N = 70$, $c_{bm} = 0.45$); bias/MSE/coverage columns are valid only within the Unadjusted/Lag-adjusted pair or within Exposure-weighted alone, because the two families target different estimands.
- Power and Type I error are the only measures directly comparable across all three Tier-1 specifications.

3.2 Tier 1 grid overview

- Figures 2 through 4 give a full-grid heatmap view of the Tier 1 results: trial design as column facets, N as row-block facets, analysis specification as the row axis within each block, blue-to-red fill scale for power.
- Each panel fixes two grid dimensions and varies a third, since no single 2D heatmap can display all four simultaneously.
- Panels A and B report all nine G1-G9 specifications (a dedicated simulation restricted to exponential DGP and Architecture B, $n_{sim} = 100$, preliminary).
- Panel C retains the original three (G1-G3), the only panel varying DGP decay form, an axis the G1-G9 extension did not re-run.
- Two features visible at a glance in Panel A that the headline table only shows cell by cell:
 - G1 and G2 track each other closely at every biomarker level within every design/ N block, differing by at most 0.03 (within Monte Carlo noise at this preliminary $n_{sim} = 100$), matching the higher-precision Section 3.4 result of numerical identity to two decimal places.
 - The Exposure-weighted family (G3, G4, and CR2 variants G8, G9) separates from G1/G2/G6/G7 only in Hybrid and OL+BDC columns; in the CO column the whole family underperforms at every biomarker level past zero.
- At the reference cell ($c_{bm} = 0.45$, $t_{1/2} = 1.0$): G4 reaches 0.48 against 0.86-0.89 for G1/G2/G6/G7 – the design-dependence of Section 3.4, now visible across all nine specifications.
- Whether this separation depends on carryover being present at all cannot be shown by Panel A alone, since it fixes $t_{1/2} = 1.0$ throughout.
- Panel B answers the above, for Hybrid and OL+BDC:
 - At $t_{1/2} = 0$ the Exposure-weighted family’s separation from G1/G2/G6/G7 disappears in both designs (all nine sit at 0.94-1.00), opening up only as half-life increases – confirming the separation is specifically a carryover effect.
 - Under CO the picture differs: G4 and G9 are already collapsed at $t_{1/2} = 0$ (0.48 and 0.47, against 0.88-0.92 for the rest), before carryover enters at all – the AIC mis-selection pathology of Block S9, predating $t_{1/2} > 0$ rather than growing from it.
- Ranking result from Panel B: G9 and G8 lead in a near-tie at Hybrid and OL+BDC reference cells (both around 0.88 at Hybrid, 0.77 to 0.74 at OL+BDC).
- G5 (Paired-difference) is the strongest non-G1/G2/CR2-variant specification at CO (0.77,

A: Effect of trial design, analysis specification, and biomarker effect on power ($t_{1/2} = 1$)

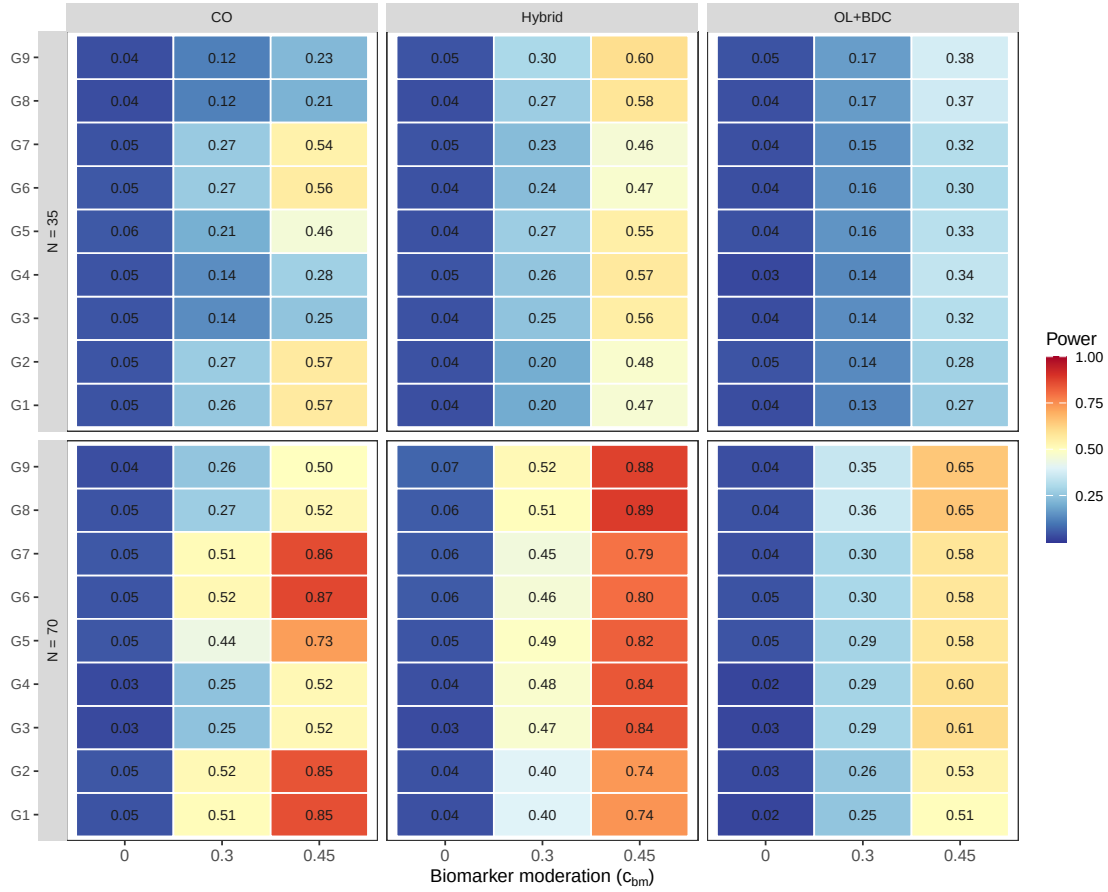


Figure 2: Tier 1 grid, biomarker-effect axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, Architecture B (preliminary $n_{sim} = 100$).

B: Effect of carryover half-life on power ($c_{bm} = 0.45$)



Figure 3: Tier 1 grid, carryover-axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, Architecture B (preliminary $n_{sim} = 100$).

versus 0.48-0.51 for the Exposure-weighted family there), consistent with G5 discarding the carryover representation that costs the Exposure-weighted family power under CO.

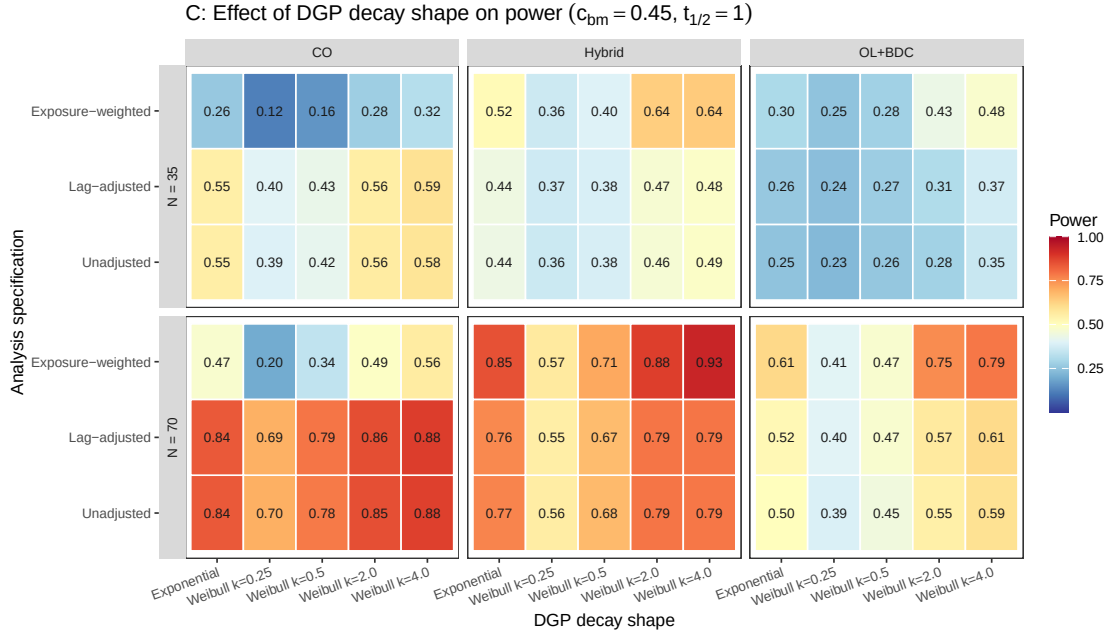


Figure 4: Tier 1 grid, decay-shape axis: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across the five DGP decay-shape levels (exponential, Weibull $k \in \{0.25, 0.5, 2.0, 4.0\}$) at $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks.

- Panel C shows the Exposure-weighted row's advantage in Hybrid and OL+BDC columns is not uniform across decay shape:
 - Widens substantially on the light-tailed side ($k = 2.0, 4.0$), reaching +0.14 to +0.21 absolute over Unadjusted/Lag-adjusted at OL+BDC.
 - Narrows on the heavy-tailed side, disappearing at the two most extreme shapes ($k = 0.25, 0.5$): statistically indistinguishable at OL+BDC (gaps of 0.00 to 0.02, well under one MCSE), barely separated at Hybrid (+0.01 to +0.04).
- This is the decay-form sensitivity of Section 3.5, shown here across the full grid rather than at one cell.

Section summary:

- The three-panel Tier 1 heatmap view confirms, across the full grid, that G1/G2 are numerically indistinguishable everywhere, and that the Exposure-weighted family's advantage is confined to Hybrid/ OL+BDC designs, growing with carryover severity, and dependent on decay-shape assumptions.
- CO is uniformly unfavorable to the Exposure-weighted family, with G4/G9 showing an AIC mis-selection pathology visible even without carryover ($t_{1/2} = 0$).

3.3 Matched-decay baseline

- At the matched cell (exponential DGP, exposure-weighted analysis, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines:
 - 0.988 ± 0.005 at $t_{1/2} = 0$.
 - 0.948 ± 0.010 at $t_{1/2} = 0.5$.
 - 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks.
 - (point estimate $\pm$ MCSE, $n_{sim} = 500$; all 500 replicates converged at every half-life in this cell)
- A previously reported comparison to Hendrickson et al.'s power of 0.82 for a comparable cell² is withdrawn.
- Reason: in the reference implementation, analysis-side and data-generating carryover parameters are set independently; the drivers reproducing the original figures supply a half-life only to the data-generating step, leaving the analysis-side half-life at its default of zero.
- At a zero half-life, the exposure-decayed predictor collapses onto the binary on-drug indicator, so the reference analysis is actually Unadjusted, not Exposure-weighted.
- The appropriate comparator is therefore the unadjusted power at the same cell, 0.766.
- The grids also differ further on interaction strength and half-life, so “the same cell” was approximate in any case.
- What the published figure was computed under has not been resolved; no claim of agreement or disagreement is made.
- This does not affect the rest of the manuscript, since analysis specifications are compared against one another within a single simulation under common random numbers.

Section summary:

- A previously claimed numerical match to Hendrickson et al.'s published power (0.82) is withdrawn: the reference implementation's half-life default of zero means the true comparator is the Unadjusted power (0.766), not Exposure-weighted.
- This unresolved external-comparison issue does not affect any internal comparison in the manuscript.

3.4 Power by analysis specification

- Two features of Figure 5 carry the substance of this study:
 - Unadjusted and Lag-adjusted curves lie on top of one another in every panel and every half-life: 0.766 and 0.770 at $t_{1/2} = 1.0$ weeks at the reference cell; across all 216 cells the mean difference is 0.003, with mean bias, MSE, coverage agreeing to three decimal places. This is read as a clean null: the two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term (the lagged-treatment term contributes nothing, Section 4.1).
 - The three specifications give nearly identical power at $t_{1/2} = 0$ (no carryover), diverging as half-life increases, with Exposure-weighted separating from the other two.

Power vs carryover under matched exponential DGP

Covariance-moderation architecture, $N = 70$, $c_{bm} = 0.45$, 500 reps/cell

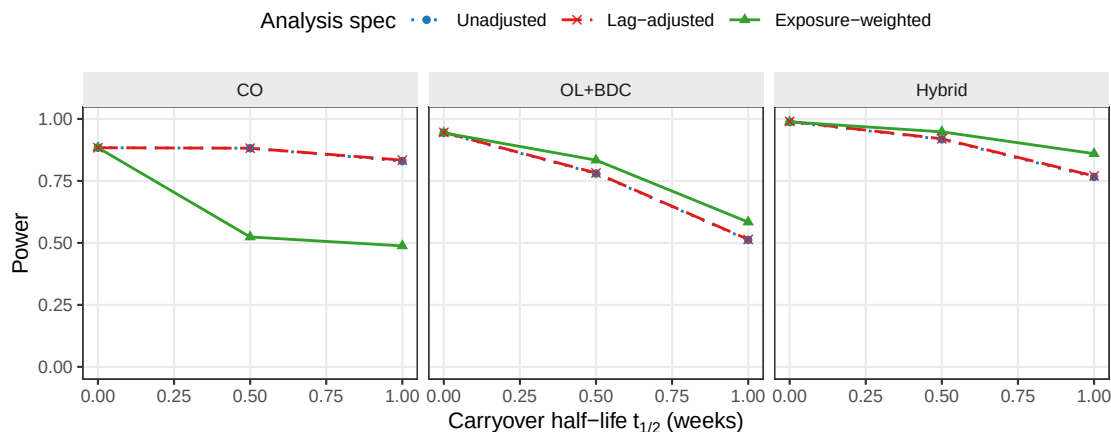


Figure 5: Power versus carryover half-life under the exponential DGP, by analysis specification and trial design.

- Exposure-weighted retains a modest but consistent advantage in Hybrid and OL+BDC designs from $t_{1/2} = 0.5$ weeks onward.
- Under crossover, Exposure-weighted is markedly inferior to both alternatives (0.488 against 0.830 for Unadjusted).
- Proposed mechanism: the within-subject correlation in a crossover trial already carries most of the signal the decaying predictor targets, so it gains little there while its down-weighting of off-drug observations works against it (Section 4).

Section summary:

- Unadjusted and Lag-adjusted are statistically indistinguishable at every half-life and every design, the manuscript's cleanest null result.
- Exposure-weighted's advantage over the other two grows with half-life in Hybrid/OL+BDC but reverses sharply under crossover (0.488 vs 0.830), attributed to crossover's within-subject correlation already capturing the signal the decayed predictor targets.

3.5 Decay-form mis-specification

- The exponential-DGP/Exposure-weighted cell is the matched reference, power 0.850 ± 0.016 in this figure's own (dedicated decay-shape) simulation; the 0.860 reported for the nominally same cell in Table 2 and Figure 5 comes from an independent draw of the same DGP, agreeing within one MCSE.
- Unadjusted and Lag-adjusted columns, invariant to assumed decay form by construction, are near-identical throughout, so variation down those columns reflects data-generating decay and Monte Carlo noise only.
- The Exposure-weighted row gives the cost of assuming exponential decay when the truth is Weibull, and it is asymmetric:

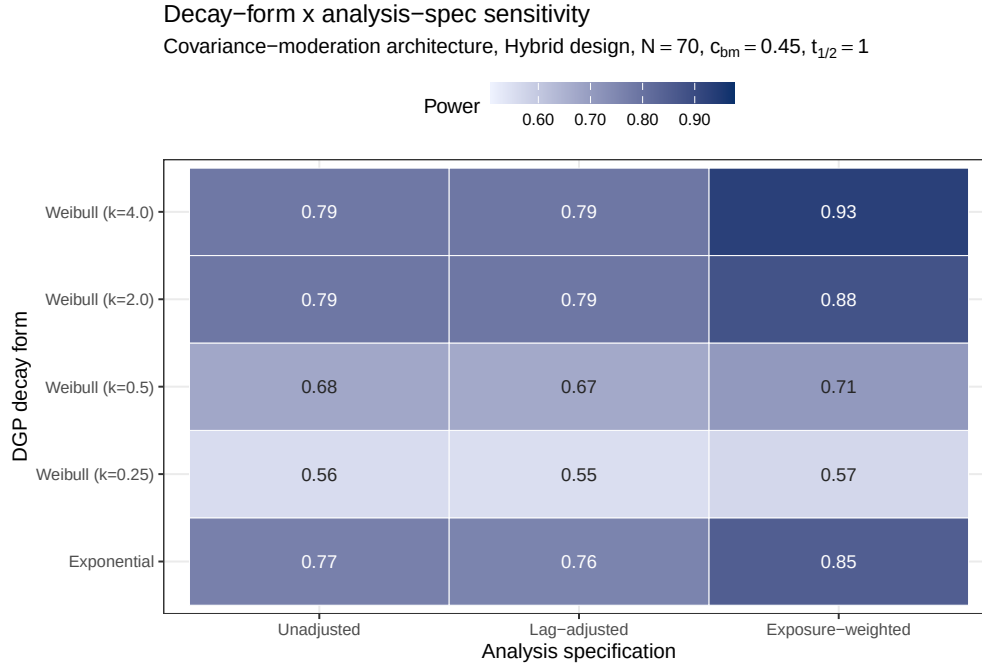


Figure 6: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, $c_{bm} = 0.45$) across five DGP decay forms (rows) and specifications (columns). Fill scale is stretched to the observed range, not $[0,1]$, and not comparable to other figures.

- Light-tailed side ($k = 2.0$): 0.880 against 0.788 for both alternatives (+0.092).
- Light-tailed side ($k = 4.0$): 0.930 against 0.786/0.788 (+0.142 to +0.144).
- Heavy-tailed side ($k = 0.5$): 0.708 against 0.676 and 0.672 (+0.032 to +0.036, roughly 1.5 MCSE).
- Heavy-tailed side ($k = 0.25$, most extreme): 0.568 against 0.558 and 0.554 (+0.010 to +0.014, well under one MCSE).
- Interpretation: the advantage erodes to nothing under a sufficiently heavy-tailed mis-specification, rather than reversing (Section 4).

Section summary:

- Exposure-weighted's advantage under exponential-decay assumption widens on the light-tailed Weibull side (up to +0.144 at $k = 4.0$) and shrinks toward zero on the heavy-tailed side, effectively vanishing at the two most extreme shapes examined.

3.6 Type I error control and cluster-robust recalibration

- Empirical Type I error at the $c_{bm} = 0$ cells of Figure 2 sits uniformly below nominal 0.05 for all nine specifications, across every design and N examined.
- This is not sampling noise but the model-based test's conservatism (Section 2.7), from a working covariance that does not match the data-generating one.
- Because the conservatism is uniform across specifications, it does not disturb the comparison among them; the cluster-robust recalibration below confirms CR2 restores nominal

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, 3,000 replicates). κ is the calibration ratio defined in Section 2.7. $\kappa > 1$ indicates a conservative test. All three specifications are conservative under the model-based test, most so Exposure-weighted ($\kappa = 1.10$, Type I = 0.029). The CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for all three.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 4: Cluster-robust recalibration at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$). EW is Exposure-weighted, LA is Lag-adjusted, U is Unadjusted. κ is the calibration ratio of Section 2.7, here for Exposure-weighted. A value above one is a conservative test. The two gap pairs give each alternative against the unadjusted benchmark under each inference. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on Exposure-weighted. Computed with `clubSandwich`.

Cell	κ model	κ CR2	EW model	EW CR2	EW-U model	EW-U CR2	LA-U model	LA-U CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	-0.003	+0.002	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	-0.002	+0.000	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	+0.000	-0.004	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	-0.003	+0.002	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	-0.004	+0.010	5

- rejection without changing which specification leads.
- Also establishes that the Section 3.4 null result is not an artifact of differential test size, since Unadjusted and Lag-adjusted are conservative to the same degree.
 - Table 3 makes this explicit at the null cell, at a higher replicate count (3,000) than the Tier 1 grid affords.
 - Table 4 confirms this directly: model-based rejection rate is below nominal across all five S6 cells and all three specifications, while CR2 restores it to within the Monte Carlo band around 0.05.
 - In information-rich cells, κ runs 1.07 to 1.26 under the model-based test, restored to about one under CR2, recovering two to five points of power.
 - The Exposure-weighted advantage against the unadjusted benchmark, under model-based vs CR2 inference:
 - Reference cell: +0.080 to +0.092.
 - Small- N cell: +0.066 to +0.086.
 - Half-life mis-specification cell: +0.010 to +0.024.
 - High autocorrelation: narrows, +0.052 to +0.032 (still leads).
 - 30% MCAR dropout: small, +0.010 (read as the S3 finding restated: dropout removes the off-drug information the advantage depends on).
 - The Lag-adjusted gap is within ± 0.010 of zero at every cell under either inference, corroborating the Section 3.4 null under a second procedure and at four additional stress cells.

- This recalibration was validated only in the Hybrid, $N = 70$ cells, and should not be extended to the crossover design, where Exposure-weighted does not lead in the first place.

Section summary:

- Model-based Type I error is uniformly conservative (below nominal 0.05) across all specifications and both grid extremes; CR2 restores nominal Type I error without disturbing the specification ranking.
- CR2 preserves or slightly widens the Exposure-weighted advantage at most stress cells (reference, small N , half-life mis-spec) but narrows it under high autocorrelation and dropout; this recalibration was validated only for Hybrid at $N = 70$.

3.7 Sensitivity analyses

- Nine Tier 2 blocks (S1-S4, S7, S8, S9, S10, S11) perturb one axis at a time around a common reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding remaining factors fixed.
- Two exceptions: S9 changes the trial design itself; S11 sets $c_{bm} = 0$.
- A tenth block, S6, applies the cluster-robust correction of Section 3.6 at the same cell plus four stress cells; an eleventh, S5, is exploratory and reported only in the archived full-scope drafts.
- Blocks are reported separately and are not pooled with Tier 1 or with each other.
- Reporting depth varies by block:
 - S1, S4, S10, S11: one-paragraph summary here only, full detail in Supplement.
 - S2, S8: headline finding plus summary paragraph here, remainder of detail in Supplement.

3.7.1 S1: within-factor autocorrelation

- All nine specifications gain power monotonically with autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$.
- The Exposure-weighted advantage over Unadjusted/Lag-adjusted (roughly 0.05 to 0.08 absolute power) is preserved at every level examined.
- Suggests autocorrelation strength and specification choice act additively rather than interacting.
- Lag-adjusted tracks Unadjusted throughout (Supplement, Section S1).

3.7.2 S2: analyst-truth half-life mismatch

- Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks).
- Penalty for an incorrect assumption is modest throughout (Exposure-weighted stays within about 7 points of its correctly-specified power at every cell but one).
- Asymmetry: under-assuming the half-life is safer than over-assuming it.
- Explanation via the limiting-case relationship of Section 2.4: shrinking $\hat{t}_{1/2}$ drives the predictor toward the binary indicator (remains serviceable); inflating it drives the predic-

tor toward a constant, at which the on-off contrast and the estimand itself degenerate.

- Full cell-by-cell detail, figures, and the nine-specification extension are in the Supplement, Section S2.

3.7.3 S3: dropout

S3: Sensitivity to dropout

MCAR and MAR (severity-biased); reference cell as above

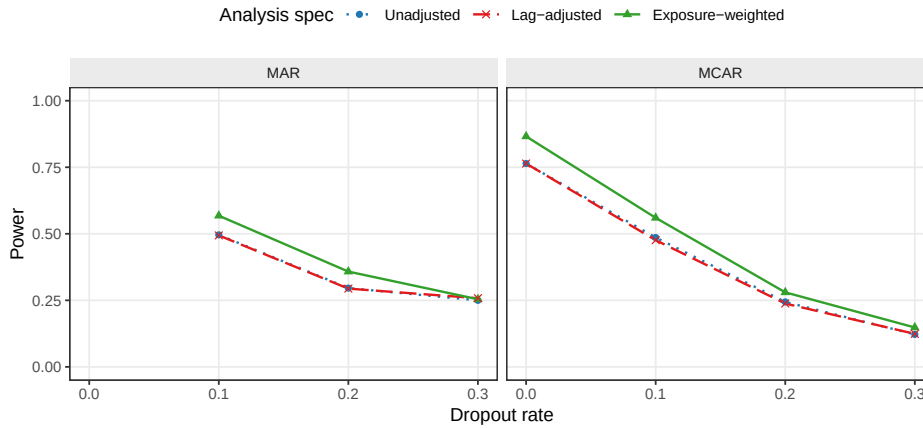


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

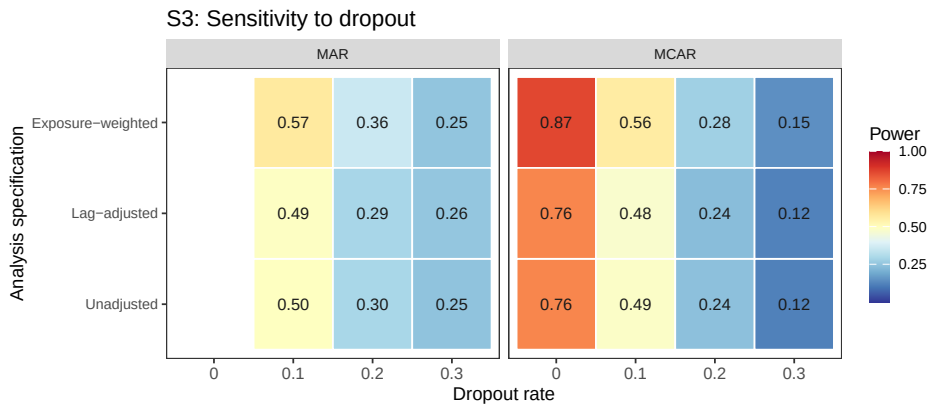


Figure 8: Block S3: power by dropout rate and analysis specification, faceted by dropout mechanism, same data as the line plot above. The MAR arm has no 0% cell (Section 3.7); it shares the MCAR reference cell.

- All three specifications lose power monotonically as dropout increases.
- At the MCAR baseline ($N=70$, no dropout), Exposure-weighted gives 0.866 ± 0.015 against 0.764 ± 0.019 for the other two.
- At 30% MCAR dropout, all three collapse to 0.122 to 0.148.
- The Exposure-weighted advantage narrows steadily as dropout worsens: about 0.10 absolute at zero dropout, 0.07 at 10%, 0.04 at 20%, 0.03 at 30% under MCAR, narrowing further under MAR at the highest rate.

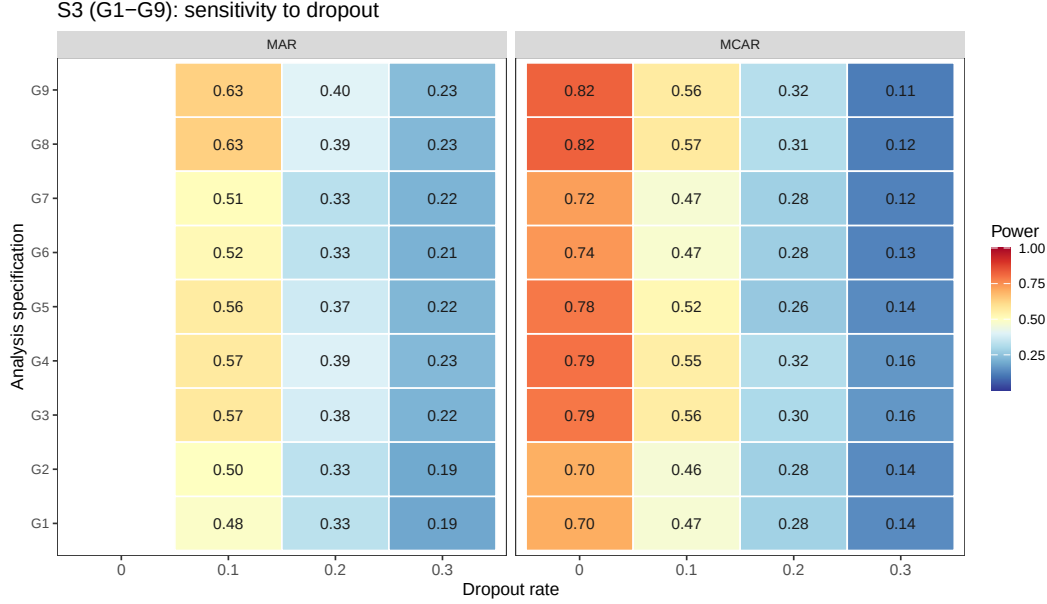


Figure 9: Block S3 extended to all nine G1-G9 specifications (preliminary $n_{sim} = 100$).

- Consistent with the advantage originating in the exposure-weighted contrast from off-drug observations, which contracts as those observations are preferentially lost.
- The advantage is preserved at dropout of 20% or below.
- Lag-adjusted again tracks Unadjusted throughout (largest discrepancy 0.010).

3.7.4 S4: biomarker-moderation effect-size curve

- The power curve is sigmoidal for all three original specifications across $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$, saturating once $c_{bm} \geq 0.45$.
- The Exposure-weighted advantage over the unadjusted benchmark is largest at intermediate effect sizes (+0.10 absolute at $c_{bm} = 0.45$), shrinking as all three approach the ceiling.
- Implication: the advantage matters most for trials targeting a moderate biomarker-treatment interaction.
- Full curve and nine-specification extension: Supplement, Section S4.

3.7.5 S7: alternative-paradigm specifications

- An earlier version of this block tested two carryover terms crossed with the biomarker:
 - E5: $S_x \sim bm + t + D_{it} + bm:D_{it} + L_{it} + bm:L_{it}$.
 - E6: $S_x \sim bm + t + D_{it} + bm:D_{it} + t_{sd} + bm:t_{sd}$.
- Motivated by Section 4.1's observation that a nuisance-main-effect carryover term cannot repair the interaction's attenuation.
- Both underperformed Unadjusted at every cell examined, evaluated by a joint 2-df Wald test (each adds a second interaction coefficient).
- Traced to a structural cause (Section 4.6): the DGP has a single true decay parameter,

Table 5: Block S7: all five G1-G5 specifications at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, $n_{sim} = 500$, all replicates converged). G4 targets bm:Dbc at the AIC-selected half-life (candidate grid 0.25, 0.5, 1.0, 2.0); G5 targets the slope of the patient-level diff bm regression. Neither is the same estimand as bm:Db (G1/G2), so only power is comparable across rows. Mean selected t1/2 is the average AIC-chosen half-life for G4 across replicates, against a true half-life of 1.0.

Specification	Power (MCSE)	Mean selected t1/2
G1: Unadjusted	0.748 (0.019)	–
G2: Lag-adjusted	0.748 (0.019)	–
G3: Exposure-weighted	0.826 (0.017)	–
G4: AIC-selected t1/2	0.832 (0.017)	1.31
G5: Paired-difference	0.804 (0.018)	–

so a second, DGP-unmotivated interaction coefficient mostly adds estimation noise.

- E5/E6 were replaced by **G4** (AIC-selected half-life) and **G5** (paired-difference regression), both single-coefficient tests, unlike E5/E6.

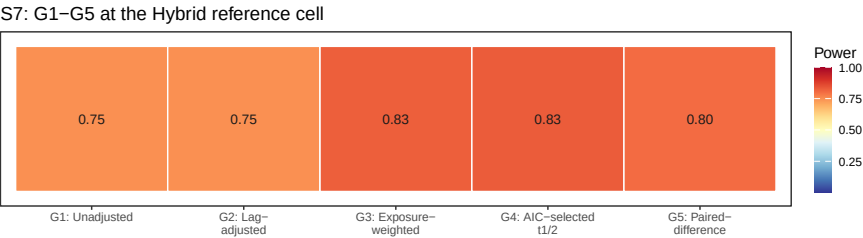


Figure 10: Block S7: power by analysis specification, same figures as the table above.

- Table 5 gives a genuinely different result from the retired E5/E6 comparison: **G4** and **G5** both beat Unadjusted at this cell, and **G4** edges out **G3** (Exposure-weighted) too.
- Ranking at this cell: **G4** leads (0.832), **G3** follows (0.826), then **G5** (0.804), then **G1/G2** tied at the back (0.748 each, the Section 3.4 null replicated once more).
- G4**’s AIC selection is imperfect but usefully informative: mean selected half-life is 1.31 weeks against a true value of 1.0, with the selection distribution concentrating on the true value and its immediate neighbor (a computed percentage choosing 1.0 or 2.0 weeks, per the inline `r` computation in the source). This is close enough to the truth that **G4** slightly exceeds **G3**’s own oracle-half-life power at this cell.
- Notable result in **G5**: it discards the repeated-measures structure, correlation model, and any carryover representation entirely, and still beats **G1/G2**, echoing Senn, Julious and Araujo’s⁷ finding that simplicity is not obviously costly for N-of-1 treatment-effect estimation.
- These results should not be read as general: Block S9, next, tests whether the ranking survives a different trial design, and it does not for every specification equally.

3.7.6 S8: architecture sensitivity of the specification ranking

- Every result reported to this point (Tier 1 and blocks S1-S7 alike) is restricted to Architecture B (covariance moderation), the manuscript's stated scope (Section 2.3).
- Block S8 asks a narrower question than a full architecture comparison (the purpose of the companion manuscript³): does the specification ranking established under Architecture B hold under Architecture A (mean moderation), or is the choice of carryover representation itself architecture-dependent?
- Refits the five S7 specifications at the same reference cell under both architectures, at three half-lives ($t_{1/2} \in \{0, 0.5, 1.0\}$).
- The S7 result (G4/G5 both beating G1) turns out to be architecture-specific, joining Exposure-weighted in a pattern that grows with carryover severity rather than switching on at a threshold.
- Under Architecture B at $t_{1/2} = 1.0$: G4 leads (0.850), ahead of G5 (0.842), G3 (0.836), and G1/G2 (0.760).
- Under Architecture A the pattern reverses for G4 and G3 alike: G1/G2 lead (0.972/0.976), G5 trails only slightly (0.964), G3 (0.946) and G4 (0.950) trail by 2 to 3 points – a real, if modest, reversal.
- Matches the archived full-scope drafts' finding that Exposure-weighted is marginally dominated under mean moderation (Section 4.6); G4 inherits the same sensitivity, sharing Exposure-weighted's formula.
- G5 is comparatively more robust to architecture, never reversing to a large disadvantage either way.
- Lag-adjusted (G2) remains statistically indistinguishable from Unadjusted (G1) under both architectures at every half-life, extending Section 3.4's clean null to Architecture A as well.
- The manuscript's one specification recommendation (exposure weighting in Hybrid/OL+BDC) is therefore conditional on Architecture B and grows more important as carryover worsens; the same qualification now extends to G4.
- Full cell-by-cell detail, table, and both architecture panels: Supplement, Section S8.

3.7.7 S9: does the reference-cell ranking survive under CO?

- Blocks S7-S8 found G4 and G5 both beat Unadjusted at the Hybrid reference cell under Architecture B, with G4 slightly exceeding Exposure-weighted (G3) itself.
- Block S9 asks whether that ranking is specific to Hybrid's short, compressed washout, refitting all five G1-G5 specifications under CO (off-drug arm spanning 2.5 to 20 weeks post-discontinuation, Section 2.2) at two true half-lives.
- It is not design-general, and including G3 changes the diagnosis reached from G4 alone.
- At $t_{1/2} = 1.0$: G1/G2 lead (0.834/0.838); G5 trails moderately (0.750); G3 and G4 both collapse, to 0.482 and 0.474 respectively, statistically indistinguishable from each other (gap 0.008, well under one MCSE).
- At $t_{1/2} = 0.5$ the same pattern holds (0.866/0.864 for G1/G2, 0.778 for G5, 0.546/0.552 for G3/G4, again indistinguishable).

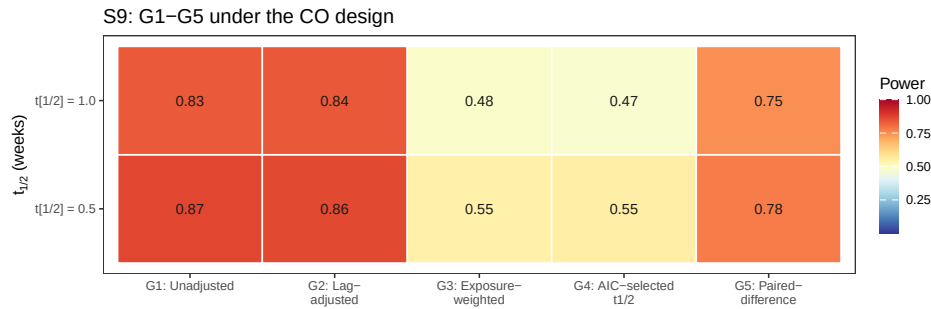


Figure 11: Block S9: power under the CO design at two true half-lives, all five G1-G5 specifications.

- Because **G3** uses the *true* half-life (not estimated), its near-identical collapse shows **G4**’s CO failure is inherited almost entirely from Exposure-weighted’s own established weakness under this design (Section 3.4: 0.488 versus 0.830), not created by the AIC-selection step.
- A separately confirmed pathology exists in the selection step itself: it concentrates overwhelmingly on the largest candidate half-life (2.0 weeks, chosen in 483 of 500 replicates at $t_{1/2} = 1.0$, 435 of 500 at $t_{1/2} = 0.5$, regardless of true half-life), consistent with a weakly informative likelihood surface when off-drug occasions are few and widely spaced.
- But that mis-selection costs **G4** essentially nothing beyond what **G3** already loses under CO with the correct half-life in hand; the dominant story is the continuous decayed predictor’s mismatch with CO’s design generally, with AIC selection an identified but practically minor addition.
- **G5**’s milder reversal has no comparable mechanical explanation: it reflects that CO’s own within-subject contrast already captures most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures structure costs more under CO than under Hybrid.
- None of **G3**, **G4**, **G5**’s Hybrid-cell behavior is design-general; the design-specific guidance in Section 4.5 should be read accordingly: **G3**/**G4** should not be used under CO regardless of whether the half-life is assumed or estimated.

3.7.8 S10: does a CR2 correction change the ranking among G1-G9?

- Section 3.6 (Block S6) showed the model-based interaction test is mildly conservative and CR2 recovers two to five points of power without disturbing the ranking among **G1**, **G3**, **G2**; that correction had not been applied to **G4** or **G5**.
- This block asks whether CR2 helps **G4** and **G1**/**G2** the way it already helps **G3**, and whether it changes which specification leads at the Hybrid, Architecture B reference cell.
- Uses the compact **G1**-**G9** naming layer: **G6**-**G9** cross CR2 with **G1**, **G2**, **G4**, **G3**. **G5** has no CR2 variant (paired-difference regression collapses to one row per patient before fitting, leaving no cluster structure for CR2 to correct).
- CR2 helps every specification it was applied to:

- G1/G2 (both 0.748 model-based) rise to G6 0.776 and G7 0.764.
- The ranking among the five base specifications is unchanged by these gains, since G1/G2 remain well below G3-G5 regardless.
- What does change the ranking: G9 – CR2 lifts G4 (0.832 model-based) to 0.870, a 3.8-point gain, the largest CR2 correction observed anywhere in this manuscript, enough for G9 to overtake every other specification tested, including G8 (0.863).
- This did not come with a Type I error cost (checked directly in Section S11 of the Supplement): every replicate converged under G9, and the power gain is consistent with CR2 correcting real conservatism rather than papering over a different problem.
- Full detail, reference-cell table, and heatmap: Supplement, Section S10.

3.7.9 S11: is G4's Type I error controlled?

- G4 selects its half-life by AIC then tests the interaction coefficient from the selected model, a post-selection inference problem: the same data drive both selection and test, which can in principle inflate Type I error beyond nominal.
- Block S11 checks this directly, refitting G1-G5 and G9 at the null ($c_{bm} = 0$) at the Hybrid reference cell, with $n_{sim} = 2000$ for tighter precision on a rate expected near 0.05.
- The concern does not materialize: G4's Type I error is 0.038, conservative and in the same range as G1-G3 (0.034-0.036), not inflated by the selection step.
- G9 (G4+CR2) is 0.052, indistinguishable from nominal 0.05.
- Read together with Section 3.6's power results: G9's status as the best-performing specification (Section 3.7, Block S10) is not an artifact of an uncontrolled test – it wins on power while holding Type I error at nominal, the same standard G3+CR2 (G8) is held to.
- This does not generalize beyond the single cell tested.
- Full detail and heatmap: Supplement, Section S11.

Section summary:

- Nine Tier 2 sensitivity blocks (S1-S4, S7-S11), plus S6 (CR2 recalibration) and an out-of-scope S5, perturb one axis at a time around the Hybrid/ $N=70/c_{bm}=0.45$ /exponential/ $t_{1/2}=1.0$ reference cell.
- Robustness holds for the core Exposure-weighted advantage under autocorrelation (S1) and moderate half-life mis-assumption (S2), narrows under dropout (S3) and at extreme effect sizes (S4).
- The alternative-paradigm specifications G4 and G5 (replacing the structurally flawed E5/E6) beat Exposure-weighted at the Hybrid Architecture-B reference cell (S7), but both results reverse or collapse under Architecture A (S8) and under the CO design (S9), so neither generalizes beyond that single cell type.
- CR2 correction helps every specification, most dramatically G4 (yielding G9, the single best-performing specification found anywhere in this manuscript), without inflating Type I error (S10, S11) – but again validated at only one cell.

3.8 A higher-precision check

- Because a number of the Tier 1 gaps are small, a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) was rerun at 600 trials per cell, all converging.
- Specifications were compared with a paired McNemar test, considerably more sensitive than an independent-groups comparison since all three are fit to the same trials.
- Type I error across the twelve null cells fell in $[0.003, 0.053]$, so nominal alpha holds.
- Paired McNemar tests place Exposure-weighted ahead of the unadjusted benchmark at all four alternative cells, by +2.3 to +13.7 percentage points.
- Largest gap at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $p = 6.6 \times 10^{-17}$), with $p \leq 0.015$ at the remaining three – a sharper confirmation than the Tier 1 grid alone provides.
- One scope caveat governs the rerun’s third arm: it was produced by the package’s `lme_analysis()` routine, whose carryover option appends time since discontinuation as a linear main effect rather than the lagged indicator L_{it} used in the Tier 1 grid; it is therefore **not** the Lag-adjusted specification of Section 2.4 and is not reported as such here.
- That alternative remains a reasonable specification not yet evaluated (Section 4.6).

Section summary:

- A higher-precision, paired-McNemar rerun (600 trials/cell, 24 cells, OL+BDC only) confirms Exposure-weighted’s advantage with much sharper statistical significance than the Tier 1 grid alone, while flagging that its third-arm comparator differs subtly from the manuscript’s own Lag-adjusted definition.

4 Discussion

4.1 The lagged-treatment term does no work

- The clearest result of this study is a negative one: adding the classical lagged-treatment indicator changed nothing measurable, anywhere in the grid.
- Across all 216 cells, mean power difference against the unadjusted benchmark was 0.003 against an MCSE of 0.018; bias, MSE, coverage agreed to three decimal places, in every sensitivity block and all five cluster-robust stress cells under both inference procedures.
- The strength of this conclusion rests on the design rather than grid size: the two specifications estimate the same coefficient, are fitted to the same simulated datasets under common random numbers, and differ by exactly one term.
- No confounding from a change of estimand, Monte Carlo variation between arms, or differential test size (both conservative to the same degree, Section 3.6); what is left is the term itself, and it contributed nothing.
- Mechanism follows from where the term sits: carryover damages the interaction test on two fronts (Section 1) – attenuates the coefficient (contaminated occasions coded as drug-free) and inflates residual variance (heterogeneity the model does not represent).
- The lagged indicator enters only as a nuisance main effect with no $\text{bm}:L_{it}$ product, so

the tested interaction remains $\text{bm}:D_{it}$ and the attenuation is untouched.

- It can in principle absorb some of the residual-variance effect, but evidently absorbs too little to matter: it marks a single occasion per randomization path and is constant thereafter, unable to describe a graded decline across a two- to four-occasion washout.
- A targeted check on a more capable, elapsed-time-indexed nuisance term ($\text{Sx} \sim \text{bm} + \text{t} + \text{Db} + \text{bm}:\text{Db} + \text{tsd}$, the “washout-adjusted” specification E4 from the reference implementation) was run across the three designs at true half-lives 0.5 and 1.0 weeks ($N = 70$, $c_{bm} = 0.45$, exponential DGP, 500 replicates):
 - Type I error at nominal for both (mean 0.050 vs 0.045).
 - Power exceeds the benchmark in all six cells by a mean of 1.3 points (sign test $p = 0.031$).
 - Reaches significance under crossover at both half-lives (+1.8, $p = 0.008$; +2.2, $p = 0.003$) and OL+BDC at the longer half-life (+2.2, $p = 0.015$), but not Hybrid (+1.0, $p = 0.27$).
 - Mechanism matches Section 2.4’s prediction: point estimates are unchanged (nothing about attenuation is repaired), the entire effect runs through the standard error, which only a long enough washout gives it something to describe.
 - Crossover places off-drug occasions 2.5 to 10 weeks post- discontinuation, against one to two weeks for Hybrid.
 - Practical ceiling on any carryover nuisance term is therefore about two percentage points, only in long-washout designs, against the nine to eleven points exposure weighting delivers where it works.
 - This check covers only the exponential DGP at a single N and effect size, so it is a probe rather than a fourth arm; it shares every specification’s structural limitation of entering carryover as a main effect only (Section 4.6).
- Practical implication: a carryover nuisance term of either form should not be expected to recover meaningful power lost to carryover, and reporting one does not constitute an effective adjustment for it.

Section summary:

- Both the classical lagged-treatment term and a more capable elapsed-time nuisance term (E4) fail to meaningfully repair the interaction test’s power loss from carryover, because a nuisance main effect cannot touch the attenuation mechanism, only (weakly) the residual-variance mechanism.
- The practical ceiling on any nuisance-term-only carryover adjustment is about two percentage points, in long-washout designs only, versus the nine-to-eleven-point gain from exposure weighting where it works.

4.2 The exposure-weighted advantage, and where it fails

- At its best cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$), the exposure-weighted predictor exceeds the unadjusted benchmark by about 10 points of power (0.860 versus 0.766, roughly five MCSEs, not attributable to chance), confirmed at all four cells of the Section 3.8 high-precision rerun.

- Under crossover it performs considerably worse (0.488 versus 0.830).
- Proposed mechanism: the within-subject correlation under crossover already carries most of the signal the decaying predictor was designed to recover, so it gains little there while its down-weighting of off-drug observations works against it.
- The recommendation therefore holds only in the two non-crossover designs and should not be extended to a crossover trial, where the unadjusted binary indicator is materially better.
- That the one specification addressing attenuation (Exposure-weighted) is also the only one to gain power, while the one addressing only residual variance (Lag-adjusted) gains nothing, suggests attenuation is the mechanism that matters here – read as an interpretation consistent with the results, not a finding the design isolates directly.

Section summary:

- Exposure weighting’s power advantage is substantial (~10 points) in Hybrid/OL+BDC designs but reverses under crossover, attributed to crossover’s own within-subject correlation already capturing the signal the predictor targets.
- The pattern (only the attenuation-addressing specification gains power) is offered as a consistent interpretation, not a directly isolated causal finding.

4.3 What the robustness checks add

- The Section 3.7 sensitivity blocks show the exposure-weighted advantage, where it exists, is not uniformly fragile but not unconditionally robust either:
 - Under incorrect decay-shape assumptions (Figure 6, Section 3.5), it widens on the light-tailed side and narrows to statistical insignificance on the heavy-tailed side, disappearing entirely at the two most extreme Weibull shapes examined ($k = 0.25, 0.5$).
 - It survives an incorrectly assumed half-life, short of a large over-assumption at a short true half-life (Block S2, Supplement Section S2).
 - It holds its five- to eight-point lead across the autocorrelation range examined (Block S1, Supplement Section S1).
- Decay-shape mis-specification is therefore the one robustness axis examined here that can erode the advantage away entirely, rather than only narrowing it.
- One caveat governs the Tier 1 figures throughout: outside Block S2, the analysis-side half-life is set equal to the data-generating one, granting Exposure-weighted an oracle no real analyst possesses, so its Tier 1 margins are upper bounds, and Block S2 gives the more realistic figure.

Section summary:

- Robustness of the exposure-weighted advantage varies by axis: robust to autocorrelation and moderate half-life mis-assumption, fragile to decay-shape mis-specification (can vanish entirely at extreme shapes).
- All Tier 1 figures outside Block S2 grant Exposure-weighted an oracle half-life, so those margins should be read as upper bounds.

4.4 Dropout matters more than the method

- The finding with the largest practical consequence concerns missing data, not carryover: when patients drop out (Figure 7), power collapses for all three specifications together, to between 0.12 and 0.25 at 30% dropout, essentially indistinguishable.
- No specification offers protection, since each can only organize whatever data remain.
- MAR dropout is somewhat worse than MCAR, because the sickest patients, who carry the most signal, tend to leave first.
- Retaining patients in the trial yields considerably more power than any choice among the three specifications ever could.

Section summary:

- Dropout is the single largest power lever in the study, dwarfing any effect of analysis-specification choice; no specification offers protection against it.

4.5 What analysis is best in which situations, in practice

Collecting the findings above into a decision rule rather than a set of pairwise comparisons:

- **Covariance moderation (Architecture B), Hybrid or OL+BDC design.** Use Exposure-weighted with a CR2 cluster-robust standard error.
 - The advantage holds under half-life mis-specification short of a large over-assumption (Section 3.7, Block S2), and across the autocorrelation range examined (Block S1).
 - It holds and widens under decay-shape mis-specification on the light-tailed side, but narrows to statistical insignificance at the two most extreme heavy-tailed shapes examined (Section 3.5): this recommendation should not be read as holding under an arbitrarily heavy-tailed decay form.
 - CR2 restores nominal Type I error without disturbing the ranking (Block S6, validated at $N \in \{35, 70\}$ in the Hybrid design specifically; OL+BDC was not itself stress-tested under CR2).
 - This remains the recommended default: it has been through the full robustness battery G4/G5 have not (next point).
- **G4 (AIC-selected half-life) and G5 (paired-difference) are promising at this same cell, not yet validated defaults.**
 - Both beat Exposure-weighted itself at the Hybrid, Architecture B reference cell (Blocks S7-S8: G4 0.850, G5 0.842, G3 0.836, at $t_{1/2} = 1.0$), and G4 in particular removes the need to assume a half-life at all.
 - Neither has been tested against decay-shape mis-specification, dropout, or under CR2 the way G3 has, so this is grounds to investigate them further, not to displace the established default yet.
 - Adding a CR2 correction to G4 (G9 in the nine-way naming layer) goes further still: at this same cell it outperforms every one of the other eight specifications, including G3+CR2, with no accompanying Type I error inflation (Section 3.7, Blocks S10-S11).

- This is the single best result anywhere in this manuscript, and also the least validated: it rests on one cell, with none of G4’s own robustness gaps yet closed.
- **Crossover (CO) design, any architecture.** Use Unadjusted.
 - Exposure-weighted is markedly worse under CO (Section 3.4), and Block S9 shows G4 and G5 fail here too, G4 catastrophically.
 - G4’s AIC half-life selection systematically drifts to the largest candidate in its grid under CO’s sparse, widely spaced off-drug sampling (chosen in over 85% of replicates regardless of true half-life), collapsing its power to roughly half of Unadjusted’s; it should specifically be avoided under any design with few, widely spaced off-drug observations.
 - G5 also underperforms Unadjusted under CO, though far more mildly.
 - The classical lagged-treatment remedy adds nothing here either, including in the one design where its washout is long enough to give it a chance (Section 4.1).
- **Architecture uncertain, or known to be mean moderation (Architecture A).** Use Unadjusted or Lag-adjusted, not Exposure-weighted or G4.
 - Both specifications’ advantage under covariance moderation reverses, marginally, under mean moderation (Block S8).
 - G5 is the one specification whose performance does not depend much on which architecture generated the data (small under B, smaller under A), but it has been tested at only one design and reference cell outside the Hybrid comparison (Block S9).
- **Never rely on Lag-adjusted as a carryover remedy.**
 - It is statistically indistinguishable from Unadjusted everywhere tested in this manuscript, Tier 1 and every Tier 2 block alike (Section 4.1).
 - If a reviewer or protocol expects some carryover term to be present, including it costs nothing measurable, but it should not be reported as having adjusted for carryover.
- **Minimize dropout before optimizing specification choice.**
 - It is the single largest lever in the entire grid (Section 4.4), and no specification, including G4/G5, has been shown to offer any protection against it; their behavior under dropout is untested.

4.5.1 Why G4 and G5 succeed at Hybrid but fail under CO

- Both new specifications’ Hybrid, Architecture B advantage evaporates under the CO design (Block S9), for two different, design-specific reasons rather than a shared structural flaw of the kind that sank E5/E6 (Section 4.6).
- G4’s failure is mechanical: with only a few, widely spaced off-drug occasions, the likelihood surface across candidate half-lives is weakly informative, and AIC selection drifts to the boundary of its candidate grid rather than locating the true value, systematically overestimating the half-life.
- Section 3.7’s S2 block already showed over-assumption is the more damaging direction of half-life error when the half-life is fixed by the analyst; under CO, G4 reproduces that same damaging error through the estimation procedure itself, on nearly every replicate.

- Concrete, testable prediction: **G4** should be reliable whenever a design gives several off-drug occasions across a graded range of elapsed time (as Hybrid does despite its short overall washout), and unreliable whenever off-drug sampling is sparse, however long the washout runs in calendar time.
- **G5**'s failure has no comparable mechanical account and is milder: the most plausible explanation is that CO's own within-subject contrast (Section 3.4) already carries most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures and correlation structure costs more under CO than under Hybrid, where that structure is doing more work. This was not tested directly and would require comparing **G5** against the within-cell decomposition of **G3**'s advantage to confirm.
- Two implications follow:
 - These results reinforce rather than undercut the earlier **E5/E6** diagnosis (Section 4.6): the more fundamental unresolved question is not which design an alternative specification is tested under, but whether the underlying DGP rewards that specification's added structure at all – neither **G4** nor **G5** was tested against a DGP built to favor them specifically.
 - The practical recommendation is narrower than “estimate the half-life” or “use a simpler method”: both **G4** and **G5** are candidates worth validating further within the Hybrid/OL+BDC, Architecture B cell where this manuscript's other recommendations already concentrate, not general-purpose replacements for Exposure-weighted or Unadjusted.

Section summary:

- The manuscript's practical decision rule is design- and architecture-conditional: Exposure-weighted+CR2 for Hybrid/OL+BDC under Architecture B; Unadjusted for crossover or uncertain/ Architecture A; Lag-adjusted never recommended as a standalone remedy; dropout minimization takes priority over specification choice.
- **G4/G9** and **G5** are flagged as promising but under-validated, with **G4**'s CO failure traced to a mechanical AIC boundary-drift problem under sparse off-drug sampling, and **G5**'s milder CO failure attributed (untested) to CO's within-subject contrast already capturing the signal the paired-difference approach targets.

4.6 Scope, related work, and limits

- This manuscript deliberately narrows the full-scope grid to the covariance-moderation architecture and non-linear decay forms, in the interest of a self-contained evaluation of manageable length.
- The archived full-scope drafts (`archive/report.Rmd`, `archive/report-slim.Rmd`) show the exposure-weighted advantage documented here does not carry over to mean moderation, where Exposure-weighted is marginally dominated in every design examined; readers needing that cross-architecture comparison should consult those drafts rather than extrapolate from this one.
- Establishing what the originally published power values (Section 3.3) were computed

under remains outstanding work.

- The crossover tradition surveyed by Jones and Kenward⁴ would lead one to expect a shape-free lagged term to outperform a parametric predictor once the decay shape is mis-specified; results here do not support that expectation over the range examined, for two reasons:
 - The exposure-weighted penalty for an incorrect half-life is too small to be overtaken (Section 3.7).
 - More fundamentally, the lagged term does not improve on ignoring carryover at all (Section 4.1), leaving no advantage for decay-shape mis-specification to erode.
- One adjacent specification bounds the negative result of Section 4.1: every carryover term considered there (lagged and washout-adjusted alike) enters as a nuisance main effect with no product against the biomarker, so the finding is that a carryover *main effect* does not recover the attenuated signal, not that no shape-free specification can.
- The natural candidate not yet tested is a carryover term interacted with the biomarker, either $Sx \sim bm + t + Db + bm:Db + L + bm:L$ or its washout-adjusted counterpart with $bm:tsd$, letting the biomarker effect decline across the washout without committing to a rate.
- An earlier version of Block S7 tested exactly these two specifications and found a clean negative result at every cell examined, traced to a structural cause: the DGP has a single true decay parameter, so a second, interacted carryover coefficient adds estimation noise without a corresponding signal to recover.
- They were replaced by **G4** (AIC-selected half-life) and **G5** (a paired-difference regression with no carryover representation at all), addressing the same open problem (unknown decay half-life) from different angles.
- Both beat Unadjusted at the Hybrid, Architecture B reference cell, **G4** even edging out Exposure-weighted itself (Block S7); neither survives a change of trial design or architecture (Blocks S8-S9).
- **G4**'s AIC selection fails specifically under sparse off-drug sampling (Section 4.5); **G5**'s advantage appears to depend on the repeated-measures structure it discards being otherwise underused, which is design-dependent.
- Block S8 shows the Lag-adjusted null of Section 3.4 replicates under Architecture A, so it is not an artifact specific to covariance moderation. Exposure-weighted's advantage, and now **G4**'s, are architecture-sensitive, reversing in direction (though not substantially) under Architecture A (Section 3.7, Block S8).
- Three further N-of-1 analysis approaches sit outside the G1-G9 family and are not evaluated here, each requiring substantial independent methodological development rather than a drop-in specification within the existing pipeline:
 - Tang and Landes¹⁰: formula-based *t*-tests correcting a single-subject paired-difference test's SE and degrees of freedom for an estimated first-order serial correlation – the natural refinement **G5** lacks (**G5** discards within-patient correlation entirely). Their tests have no biomarker-moderation term, and would need a $diff \sim bm$ extension with a serial-correlation-aware variance to enter this comparison.
 - Schmid and Yang¹¹: a Bayesian hierarchical alternative to the frequentist

1288 mixed-model/sandwich-estimator family spanning **G1-G9**, with partial pooling of
 1289 individual- level treatment-effect, trend, and autocorrelation parameters across
 1290 patients; they note carryover is tractable only in aggregated, multi-individual
 1291 analyses like this one, but a biomarker-moderation extension of their model is a
 1292 separate undertaking.

- 1293 – Daza¹²: frames N-of-1 inference as a counterfactual, time-varying-treatment prob-
 1294 lem, estimated via g-computation or inverse-probability weighting – a structurally
 1295 different paradigm from **G1-G9**’s explicit exponential-decay or lag covariates, treat-
 1296 ing carryover as confounding to be controlled rather than a predictor to be modeled.
- 1297 • None of the three is a candidate for a tenth specification within this manuscript’s scope,
 1298 but each identifies a direction future work on this estimand should consider.
- 1299 • Two further limitations bound what can be concluded:
 - 1300 – The numbers presented come from a single parameter set calibrated to a prazosin
 1301 and PTSD trajectory, so absolute power values should be read as illustrative, and
 1302 the ranking (itself conditional on trial design) as the more portable result.
 - 1303 – The sensitivity checks vary one factor at a time and can therefore miss genuine in-
 1304 teractions between factors (e.g., between high autocorrelation and heavy dropout);
 1305 the one joint check available (Block S5 in the full-scope manuscript) revealed no
 1306 such interaction.

1307 Section summary:

- 1308 • This manuscript is a deliberately narrowed slice of a larger full-scope study; the archived
 1309 drafts remain the authority on mean moderation and should be consulted before gener-
 1310 alizing beyond covariance moderation.
- 1311 • The crossover literature’s expectation that a shape-free lagged term would outperform a
 1312 parametric predictor under decay-shape mis-specification is not supported here, because
 1313 the lagged term fails even without mis-specification.
- 1314 • Three substantial alternative N-of-1 analysis paradigms (serial- correlation-corrected
 1315 paired tests, Bayesian hierarchical models, counterfactual/causal-inference framing) re-
 1316 main outside this manuscript’s scope and are flagged as future-work directions, not
 1317 rejected findings.
- 1318 • Conclusions rest on a single calibrated parameter set and one-factor-at-a-time sensi-
 1319 tivity checks, so absolute power values are illustrative and possible factor interactions
 1320 beyond the one jointly checked (autocorrelation x dropout, no interaction found) remain
 1321 unexamined.

1322 5 Conclusions

1323 1. The lagged-treatment term does nothing.

- 1324 • It never improved on ignoring carryover altogether, on any measure, in any design,
 1325 at any half-life, and in every sensitivity block.
- 1326 • A clean null given the two specifications share an estimand and differ by a single
 1327 term.

- The term enters as a nuisance main effect only, leaving the attenuation of the interaction untouched.
 - Should not be reported as an adjustment for carryover.
2. **Exposure weighting helps, but conditionally.**
- Achieves the highest power, about 10 points over the unadjusted benchmark, only in the Hybrid and OL+BDC designs.
 - Its lead survives mis-specified half-life there, and widens under a light-tailed decay-shape mis-specification, but narrows to statistical insignificance at the two most heavy-tailed Weibull shapes examined; not robust to an arbitrarily heavy decay tail.
 - Under crossover it is markedly worse than the binary indicator (0.488 against 0.830) and should not be used there.
 - Where used, a CR2 cluster-robust standard error is recommended (Block S6).
3. **An approximate half-life is adequate, but err short.**
- A fourfold error at a true half-life of one week costs 6.6 points of power, so a rough pharmacokinetic estimate suffices in practice.
 - The error is asymmetric: under-assuming drives the predictor toward the still-serviceable binary indicator, while over-assuming erodes the advantage to a tie.
 - Outside Block S2 the analyst is granted the true half-life, so the Tier 1 margins are upper bounds.
4. **Dropout matters more than the choice of method.**
- Power falls sharply as dropout increases, roughly halving at 20%, equally across all three specifications.
 - Preventing dropout is a more effective use of design effort than any choice among them.
5. **The interaction test is somewhat conservative, and correctably so.**
- All three specifications reject slightly less often than nominal under the null.
 - A CR2 cluster-robust standard error restores the correct rate while recovering modest power.
 - The exposure-weighted advantage is held or widened under CR2 at four of five S6 stress cells (Block S6).
6. **This is a narrowed slice of a larger study.**
- Mean moderation and linear decay fall outside this manuscript's scope.
 - The archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` are the authority on those comparisons and should be consulted before generalizing beyond covariance moderation.
 - No claim of numerical agreement with the originally published power values (Section 3.3) is made.
7. **Carryover terms interacted with the biomarker do not help; two alternative-paradigm specifications do, but only at the reference cell.**
- Crossing a lagged or washout carryover term with the biomarker gave lower power than the unadjusted benchmark at every cell examined (Block S7's original version).

- Traced to a structural cause: the DGP has a single true decay parameter, so a second, interacted coefficient adds estimation noise without a corresponding signal to recover (item 1's mechanism, restated).
 - They were replaced by two specifications answering the same open question (the analyst's unknown decay half-life) differently: **G4** estimates it by AIC selection; **G5** discards the decay representation and the mixed model itself for a two-stage paired-difference regression following Senn, Julious and Araujo⁷.
 - Both beat Exposure-weighted at the Hybrid, Architecture B reference cell (Blocks S7-S8; **G4** 0.850, **G5** 0.842, **G3** 0.836 at $t_{1/2} = 1.0$).
 - Neither result generalizes: Block S8 shows both reverse, like Exposure-weighted itself, under Architecture A, and Block S9 shows both fail under the crossover design, **G5** mildly and **G4** catastrophically.
 - **G4**'s CO failure has a clear mechanism: AIC selection systematically drifts to the boundary of its candidate half-life grid when off-drug sampling is sparse and widely spaced, reproducing the S2 finding that over-assuming the half-life is the more damaging error, now through the estimation procedure itself rather than a fixed wrong assumption.
 - Both specifications should be read as promising candidates for further validation at the one cell type where this manuscript already recommends Exposure-weighted, not as general-purpose replacements for it or for Unadjusted.
8. **Adding CR2 to the AIC-selected specification gives the single best result in this manuscript, at one cell only.**
- **G9** (**G4** plus a CR2 cluster-robust standard error) outperforms all eight other specifications at the Hybrid, Architecture B reference cell, including **G3**+CR2.
 - Its Type I error holds at nominal despite the post-selection risk **G4**'s half-life selection step raises in principle (Section 3.7, Blocks S10-S11).
 - Read as a promising direction, not a recommendation: **G9** has not been tested against decay-shape mis-specification, dropout, a crossover design, or an alternative architecture, all of which item 2's specification (**G3**+CR2) has already been checked against.

Section summary:

- The eight numbered conclusions rank by evidentiary weight: items 1-6 are grid-wide, well-validated findings (lagged term does nothing, exposure weighting helps conditionally, half-life robustness, dropout dominance, CR2 correctability, scope boundaries), while items 7-8 flag **G4**/**G5**/**G9** as promising but single-cell, not-yet-generalized results.
- The manuscript's one unconditional recommendation (CR2-corrected exposure weighting in non-crossover designs under covariance moderation) is explicitly bounded by architecture and design; no specification is recommended as universally best.

6 Conflict of interest

- The authors declare no conflicts of interest.

7 Funding

- This work received no specific funding.

8 Data availability statement

- The data and code supporting the findings of this study are available in the pmsimstats-ng project repository.
- Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`.
- Simulation drivers are at `analysis/scripts/carryover-sensitivity/`.
- Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

- Manuscript rendered with R 4.6.0 inside the project's zzcollab Docker container.
- Production simulation ran under R 4.5.3 (§3.6), with image hash in `Dockerfile` and package set in `renv.lock`.
- Simulation study uses `nlme` for the continuous-time linear-mixed analysis, `parallel` for the 8-core parameter-grid sweep, and `data.table` or `furrr/purrr` in the two simulation collections.
- No simulation code runs during knitting; only pre-computed checkpoints are loaded.
- Each driver sets a master seed (20260507) and derives per-replicate seeds from it via `parallel::mc.set.seed`.

Data and code availability.

- All simulation outputs, cell-level summaries, and driver scripts are versioned under `analysis/scripts/carryover-sensitivity/` and `analysis/data/` in the pmsimstats-ng repository.
- This includes: the principal factorial; the S1-S5 sensitivity blocks; the S6 cluster-robust recalibration; the washout-adjusted (E4) check of Section 4.1; the high-precision prototype rerun of Section 3.7.
- Also includes:
 - S7 alternative-paradigm block (`09-run-sensitivity-s7.R`, `output/09-sensitivity-s7.rds`).
 - S8 architecture-by-half-life block (`10-run-sensitivity-s8.R`, `output/10-sensitivity-s8.rds`).
 - S9 CO-design block (`14-run-sensitivity-s9.R`, `output/14-sensitivity-s9.rds`).
 - S10 CR2 comparison block (`15-run-sensitivity-s10.R`, `output/15-sensitivity-s10-g.rds`).
 - S11 Type I error block (`16-run-sensitivity-s11.R`, `output/16-sensitivity-s11.rds`).
- All five were added 2026-08-18/19 (master seed 20260415, 500 replicates per cell except S11's 2000; Hybrid reference cell except S9, which uses CO, and S11, which sets $c_{bm} = 0$).
- E5/E6 (biomarker-interacted carryover terms) were tested in an earlier version of S7-S9 and are documented in Section 4.6; they were replaced by G4 (AIC-selected half-life, reusing the candidate-grid logic of `characterize_carryover()`, `R/carryover_analysis.R`) and G5 (paired-difference regression).

- The E1/E3/E2, E7, E9 fitters, and the E1cr2/E3cr2/E7cr2 CR2 variants used by S10 (`cr2_extract()`, `fit_spec_s7`, `simulate_cell_s7`), are appended to `simulation-core.R` alongside the Tier 1 `fit_spec/fit_all_specs` pair, which they do not modify.
- `spec-labels.R` additionally defines the G1-G9 compact display-code layer (`g_code`, `g_labels`) used in S10's table and figure; G8 (G3+CR2, stored as E2cr2) is reused directly from Block S6 rather than rerun, so it does not share common random numbers with the rest of S10's comparison.
- Figure 4, and the decay-form mismatch panel (Figure 6), retain the original three specifications (G1-G3), since decay shape is the only axis those two figures vary.
- Both were regenerated from a dedicated 30-cell simulation at a more extreme Weibull shape range ($k = 0.25, 0.5, 2.0, 4.0$, extended from an initial $k = 0.5, 2.0$, itself replacing an earlier $k = 0.7, 1.5$; $k = 1.0$ is not a separate condition, since it duplicates Exponential exactly) than the shared production grid (`02-grid-summary.rds`) still uses (`23-run-decay-shape-sensitivity.R`, `02-decay-shape-sensitivity.rds`, $n_{sim} = 500$, Architecture B only).
- This is a standalone script rather than a change to `01-run-factorial.R`'s shared grid, since that grid is also consumed by the archived full-scope drafts and other manuscripts.
- Figures 2 and 3 were later expanded to all nine G1-G9 specifications, from a dedicated 30-cell simulation restricted to the exponential DGP and Architecture B (`19-run-tier1-hendrickson-g9.R`, `02-grid-summary-hendrickson-g9.rds`, preliminary $n_{sim} = 100$).
- This replaced the original three-specification version of both panels rather than supplementing it, and superseded the separate Type I error checks the manuscript previously reported at Figures 6-9 (the Tier 1 Type I boxplot/heatmap and the S6 G1-G3/G1-G9 Type I dot plot/heatmaps), since the $c_{bm} = 0$ column of Panel A now reports Type I error for all nine specifications directly.
- Those retired figures' underlying data and scripts (`03-render-figures-extra-slim.R`, `07-type1-cr2-figure-extra-slim.R`, `18-run-sensitivity-s6-g9.R`) are retained for archival purposes but no longer referenced by this document.
- The corresponding Tier 2 heatmaps (S1-S4, S6, S7, S8, S9, S10, S11) are produced by `13-hendrickson-heatmap-tier2.R` from `02-sensitivity-summary.rds` and the S7/S8/S9/S10/S11 output `.rds` files (plus `diag-s6-cr2.rds` for G8 and Table 4); these Tier 2 figures are additional to, not replacements for, the existing S1-S4 line plots and the S7/S8 kable tables.
- The grid design and Morris ADEMP pre-registration are in `simulation-grid-plan.md` in the same directory.
- Superseded wider-scope drafts are retained under `archive/`, documented in its `README.md`.

Specification labels.

- All nine specifications are stored under E-prefixed codes (E1, E2, E3, E7, E9, E1cr2, E2cr2, E3cr2, E7cr2) in every data file and script, displayed uniformly as G1-G9, or by

name, at render time only.

- The mapping is applied by `spec-labels.R`, so stored values and archived outputs are unaffected: what a reader sees as **G3** (Exposure-weighted) in the prose above is stored as **E2** in every `.rds` file this manuscript reads.
- Output archived before 2026-07-31 stores the specifications as **A1**, **A2**, **A3**.
- The migration is documented in `analysis/report/NOTATION.md`.

Section summary:

- The manuscript reads only pre-computed checkpoints at render time; all simulation drivers, their output `.rds` paths, and master seeds are catalogued here for reproduction.
- Two notational layers exist in the underlying data (**E**-codes, stored; **G1-G9**, displayed) plus a legacy **A1-A3** layer in pre-2026-07-31 archives, all mapped via `spec-labels.R` and documented in `NOTATION.md`.

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